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# A robust and contact resolving Riemann solver on unstructured mesh, Part II, ALE method

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ABSTRACT

This paper investigates solution behaviors under the strong shock interaction for moving mesh schemes based on the one-dimensional Godunov and HLLC Riemann solvers. When the grid motion velocity is close to Lagrangian one, these Godunov methods, which updates the flow parameters directly on the moving mesh without using interpolation, may suffer from numerical shock instability. In order to cure such instability, a new cell centered arbitrary Lagrangian Eulerian (ALE) algorithm is constructed for inviscid, compressible gas flows. The main feature of the algorithm is to introduce a nodal contact velocity and ensure the compatibility between edge fluxes and the nodal flow intrinsically. We establish a new two-dimensional Riemann solver based on the HLLC method (denoted by the ALE HLLC-2D). The solver relaxes the condition that the contact pressures must be the same in the traditional HLLC solver and constructs discontinuous fluxes along each sampling direction of the similarity solution. The two-dimensional contact velocity of the grid node is determined via enforcing conservation of mass, momentum and total energy. The resulting ALE scheme has a node instead of grid edge conservation properties. Numerical tests are presented to demonstrate the robustness and accuracy of this new solver. Due to the multi-dimensional information introduced and consistency between the fluxes and nodal contact velocity, the developed ALE algorithm performs well on both quadrilateral and triangular grids and reduces numerical shock instability phenomena.



## 1. Introduction

In many computational fluid dynamics (CFD) applications, boundaries of the physical domain of the flow might move in time. Typical examples include flows in airfoil oscillations, wing flutter, and a large class of multi-material interface problems where the material interface is explicitly tracked as a Lagrangian surface. When moving boundaries experience large displacements, or when they undergo large deformations, the arbitrary Lagrangian Eulerian method (ALE) is a useful tool to solve the flow problems on a moving and possibly deforming grid.

Usually, ALE methods can be implemented by three approaches. The first one, which is termed as Lagrange plus remap algorithm, is proposed by Hirt et al. [24]. It includes an explicit Lagrangian step, a rezoning step and a remapping step.

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Such a method is suitable for the problems in which the Lagrangian grid and rezoned grid are independent and even have different topologies. Recently such methods have been used in multi-material flows calculation with interface reconstruction or reconnection-based technique [21,22,33]. The second is Euler plus remap algorithm, which are used in extensive adaptive moving mesh literatures [14,31,32,47,48]. The last one is named direct ALE and consists in an unsplitting moving mesh discretization of the gas dynamics equations wherein grid topologies keep unchange and convective terms are solved directly, see [3,34,41].

The early study in this direction, as we know, can be found in [28] and [23], who proposed a method of moving mesh at an adaptive speed at each time step to improve the resolution of shocks and contact discontinuities. Then, many other moving mesh methods for hyperbolic problems have been studied in the literatures, and most of these works focus on grid moving strategies.

Due to the fact that the Lagrangian mesh reflects the physical motion of the fluid, the 'rezoned' grid is required to be close to the Lagrangian one at each time step in some grid rezone strategies used in ALE computation. Knupp, Margolin and Shashkov [29] construct a so-called 'reference Jacobian method' based on optimization to maintain both geometric and physical qualities of grid. Galera, Maire and Breil in [21] present a multi-material ALE scheme, whose main feature lies in introducing a new mesh relaxation procedure which also keeps the rezoned grid as close as possible to the Lagrangian one. Hui et al. construct 'the generalized Lagrangian method' in computational space [26] and introduce a unified coordinate system [27], where the main idea of the algorithm is to move grids with the same direction of the fluid movement but with different magnitude of fluid velocity.

Adaptive moving mesh is another well-known grid rezone strategy. Locally clustering mesh points in the regions where there are large gradients in the physical quantities will effectively reduce possible errors or oscillations. Up to now, much progress has been made in adaptive moving mesh methods for the numerical solution of partial differential equations, including grid redistribution approach based on the variational principle proposed by Azarenok et al. [3,4], Brackbill [10], Yanenko et al. [54] and Winslow [51]; moving mesh PDEs methods proposed by Huang [25]; and moving mesh methods based on the harmonic mapping proposed by Chen et al. [14], Dvinsky [19], Li et al. [31,32], and Tang et al. [47,48].

Some researchers have paid more attention to numerical schemes for flow problems. Zabrodin et al. [56] design a kind of ALE method on curvilinear grid. Azarenok et al. [3] discuss a second order ALE Godunov method on moving mesh. Farhat et al. [20] investigate the influence of geometric conservation law to nonlinear stabilities of ALE scheme. Luo, Baum and Lohner [34] implement AUSM+, HLLC, and Godunov schemes in the context of ALE formulation, and find that ALE AUSM+ scheme could lead to collapse of a calculation for a class of underwater explosion problems, whereas the ALE HLLC and Godunov schemes are able to offer accurate and robust solutions for capturing strong shock and contact discontinuities. Pakmor et al. [43] find that calculating an MHD problem turns out to be unstable when the grid velocities are close to the fluid velocity and then propose a modification technique.

This paper focuses on numerical method of moving grid and reports a numerical phenomenon: under the strong shock interaction, the Godunov and HLLC methods in ALE formulation in [34] also suffer from instability on some well-used moving grids. Such unstable phenomenon comes from a genuine two-dimensional mechanism since it depends on grid aspect ratio and shock strength. The modification method proposed by Parkmor et al. [43] may be effective in the computation of MHD problems, but it is invalid in gas flow for hydrodynamic problems.

Moreover, we present a new cell-centered ALE method to solve the two-dimensional compressible gas dynamics equations on unstructured grids. The main work is to construct a two-dimensional Riemann solver. Different from some classical multi-dimensional Riemann solvers such as those proposed by Balsara [5–7], Boscheri et al. [8,9] and Wendroff [52], which are based on the multi-dimensional wave configurations, our method provides a numerical flux at a cell face after given inputs to a Riemann problem. The method has close relation with a set of cell centered Lagrangian ones with nodal solvers.

The pioneers of the nodal solvers, e.g., Després and Mazeran [16], and Maire et al. [36], have noticed that the flux computation was not compatible with the node displacement in the early Lagrangian CAVEAT algorithm [17]. This incompatibility may generate additional spurious components in the vertex velocity field. In order to attain the consistency between the flux discretization and vertex velocity, they design an approximate Riemann solver located at the nodes and therefore modify the calculation results quite well. From then, a series of contributions to the development of the method are performed. Burton et al. extended the seminal works of [16,36] by proposing a new nodal Riemann-like method that handles solid dynamics [11], and Morgan et al. apply their nodal solver to contact surface algorithm [40]. Carré et al. extend Després and Mazeran's method to arbitrary dimension [13]. Maire develops high-order schemes on two-dimensional Cartesian and cylindrical geometries [37,38]. More work can be found in [15,33] and [39].

The present method in this paper is a generalization from the Eulerian method in [46] to the ALE method, which can be also seen as an extension of the nodal solvers in Lagrangian framework. The main work is to construct a two-dimensional Riemann solver whose flux has one-dimensional form but includes multi-dimensional information. The algorithm consists of two stages: (1) a two-dimensional contact velocity defined at grid vertex is determined by conservation of mass, momentum and total energy; (2) an approximate Riemann solver HLLC-2D is constructed in which the vertex velocity is given. The solver relaxes the condition that the contact pressures must be the same in the HLLC solver and constructs two discontinuous fluxes in any sampling direction of the similarity solution. The robustness and the accuracy of this new scheme are validated by numerical results.

The outline of this paper is as follows. In Section 2 the governing equations in the context of an ALE formulation and their space discretization are described. The ALE and Lagrangian formulations based on the one-dimensional Godunov and HLLC

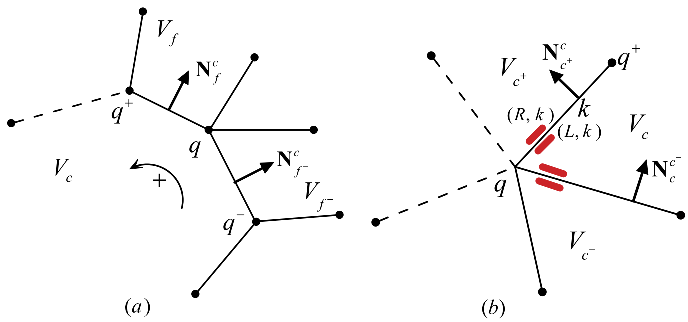


**Fig. 1.** Notations on a generic polygonal grid (a) and a generic vertex (b). The edge $c \wedge c^+$ between the cells $V_c$ and $V_{c^+}$ is sometimes denoted as $k$, and the left state and the right state are determined according to the direction of the normal vector of the edge $k$.

solvers are presented in Section 3. An unstable phenomenon is discussed in Section 4. The new ALE method is constructed in Section 5. The corresponding numerical results on both triangular and quadrilateral meshes are shown in Section 6 to provide a clear evidence of the robustness of this new scheme. Finally, the conclusions are summarized in Section 7.

## 2. Governing equations and space discretization

The governing equations for inviscid flow in two-dimensions are as follows:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = 0, \tag{1}$$

where the state vector and flux vectors are

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}, \qquad \mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ \rho E u + p u \end{bmatrix}, \qquad \mathbf{G}(\mathbf{U}) = \begin{bmatrix} \rho v \\ \rho u v \\ \rho v^2 + p \\ \rho E v + p v \end{bmatrix},$$

where $\rho$, $p$, $E$ are the fluid density, pressure and total energy respectively, and $\mathbf{u} = (u, v)$ is the fluid velocity. The equation of state is in the form

$$p = (\gamma - 1)\rho e = (\gamma - 1)\rho\left(E - \frac{1}{2}\left(u^2 + v^2\right)\right),$$

where $\gamma$ is the specific heat ratio.

It is convenient, for the subsequent discretization, to recast the system (1) of equations in the following moving control volume formulation [21,17]:

$$\frac{d}{dt}\int_{\Omega} \mathbf{U}\, dx\, dy + \int_{\partial\Omega} \left[(\mathbf{F}, \mathbf{G})\mathbf{N} - (\mathbf{w} \cdot \mathbf{N})\mathbf{U}\right] dl = 0 \tag{2}$$

where $\mathbf{w}$ is the moving velocity of the control volume $\Omega$, and $\mathbf{N}$ is the unit outward normal direction on the boundary of $\Omega$. If $\mathbf{w} = \mathbf{u}$, the system reduces to a Lagrangian formulation, and if $\mathbf{w} = 0$, it has Eulerian form.

## 3. Discrete scheme

### 3.1. Notations on a generic polygonal grid

Each cell $V_c$ of the mesh is assigned a unique index $c$. We use the index $f$ to denote a generic neighboring cell $V_f$ which has a common edge denoted as $c \wedge f$ with cell $V_c$, and edge is also denoted as $k$, refer to Fig. 1. $\mathcal{F}(c)$ is the set of such neighboring cells around cell $V_c$, and $\mathcal{Q}(c)$ is the set of all vertices of a cell $V_c$. Each vertex of the mesh is assigned a unique index $q$, and $\mathcal{C}(q)$ and $\mathcal{K}(q)$ are defined to be the sets of cells and edges respectively around the vertex $q$, i.e.,

$$\mathcal{C}(q) = \{V_c\text{: cells sharing a vertex } q\},$$

$$\mathcal{K}(q) = \{k\text{: edges sharing a vertex } q\}.$$

The physical quantities such as the density $\rho_c$, pressure $p_c$, velocity $\mathbf{u}_c$, energy $E_c$, $e_c$ are defined in the cell center of $V_c$. The moving velocity $\mathbf{w}_q$ of the grid is defined on the node $q$. In Fig. 1(a), $\mathbf{N}_f^c = ((n_x)_f^c, (n_y)_f^c)$ is the unit normal vector of cell $V_c$ on edge $c \wedge f$, where the subscript $f$ and the superscript $c$ indicate that the direction of the vector is from $V_c$ to $V_f$. According to the normal direction of an edge between two neighboring cells, sometimes we use denotation $L$ and $R$ to express the left state and right state of the edge. For example, $V_c$ and $V_f$ are the left and right cells of the edge $c \wedge f$ in Fig. 1(a), similarly, $V_c$ and $V_{c+}$ in Fig. 1(b) are denoted as $L$ and $R$ of the edge $k$.

The above equations (2) are discretized by utilizing the idea of the Godunov scheme on unstructured mesh,

$$\begin{aligned} \mathbf{U}_c^{n+1} &= \frac{|V_c^n|}{|V_c^{n+1}|}\mathbf{U}_c^n - \frac{\Delta t}{|V_c^{n+1}|}\sum_f \int_f \left(\mathcal{T}_f^c\right)^{-1}\left[\mathbf{F}\left(\mathcal{T}_f^c\mathbf{U}\right) - \left(\mathbf{w}_f^c \cdot \mathbf{N}_f^c\right)\mathcal{T}_f^c\mathbf{U}\right] dl \\ &\approx \frac{|V_c^n|}{|V_c^{n+1}|}\mathbf{U}_c^n - \frac{\Delta t}{|V_c^{n+1}|}\sum_f L_f^c\left(\mathcal{T}_f^c\right)^{-1}\mathbf{F}_f^c\left(\mathbf{w}_f^c \cdot \mathbf{N}_f^c, \mathcal{T}_f^c\mathbf{U}_c, \mathcal{T}_f^c\mathbf{U}_f\right), \end{aligned} \tag{3}$$

where the state vector with the superscript $n$ indicates it is at the time $t^n$. $|V_c^n|$ and $|V_c^{n+1}|$ denote the volume of cell $V_c$ at time $t^n$ and $t^{n+1}$ respectively, and $\Delta t = t^{n+1} - t^n$. $\mathbf{w}_f^c$ is the moving velocity of the cell edge $c \wedge f$ between $V_c$ and $V_f$, which is defined at the center of the cell edge. $L_f^c$ is the length of the edge $c \wedge f$. In this paper, all quantities appearing in flux functions $\mathbf{F}_f^c$ are set at time $t^n$ for simplicity. $\mathcal{T}_f^c$ is the rotation matrix and $(\mathcal{T}_f^c)^{-1}$ is its inverse, namely

$$\mathcal{T}_f^c = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & (n_x)_f^c & (n_y)_f^c & 0 \\ 0 & -(n_y)_f^c & (n_x)_f^c & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \left(\mathcal{T}_f^c\right)^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & (n_x)_f^c & -(n_y)_f^c & 0 \\ 0 & (n_y)_f^c & (n_x)_f^c & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{4}$$

The numerical flux $\mathbf{F}_f^c$ between the cell $V_c$ and $V_f$ can be obtained from many methods. The most popular way is to solve a one-dimensional Riemann problem along the normal direction of the edge $c \wedge f$:

$$\begin{aligned} &\mathbf{U}_t + \mathbf{F}_x = 0, \\ IVS\text{:}\quad &\mathbf{U}(x,0) = \begin{cases} \mathbf{U}_L = \mathcal{T}_f^c\mathbf{U}_c^n, & x < 0, \\ \mathbf{U}_R = \mathcal{T}_f^c\mathbf{U}_f^n, & x > 0, \end{cases} \end{aligned} \tag{5}$$

and the numerical flux can be expressed by

$$\mathbf{F}_f^c(w, \mathbf{U}_L, \mathbf{U}_R) = \mathbf{F}^w - w\mathbf{U}^w, \tag{6}$$

where $w = \mathbf{w}_f^c \cdot \mathbf{N}_f^c$ is the normal projection of the moving velocity $\mathbf{w}_f^c$ defined on the cell edge $c \wedge f$, and $\mathbf{U}^w$ and $\mathbf{F}^w$ are state and flux functions along the grid velocity $x/t = w$ respectively.

In above notations, the superscript $c$ is used to emphasize the fact that these quantities are viewed from cell $c$. From cell $f$, there are

$$L_c^f = L_f^c, \qquad \mathbf{N}_c^f = -\mathbf{N}_f^c. \tag{7}$$

### 3.2. The finite volume methods based on 1-d Riemann solvers

In most finite volume methods (e.g., [3,4,34,56]), numerical fluxes are defined at the middle point of edge $c \wedge f$, see Fig. 2(a). Usually there holds the following consistency condition,

$$\left(\mathcal{T}_c^f\right)^{-1}\mathbf{F}_f^c\left(-w, \mathcal{T}_c^f\mathbf{U}_f, \mathcal{T}_c^f\mathbf{U}_c\right) = -\left(\mathcal{T}_f^c\right)^{-1}\mathbf{F}_f^c\left(w, \mathcal{T}_f^c\mathbf{U}_c, \mathcal{T}_f^c\mathbf{U}_f\right), \tag{8}$$

and conservation condition

$$\mathbf{F}_c^f(w, \mathbf{U}_L, \mathbf{U}_R) = \mathbf{F}_f^c(w, \mathbf{U}_L, \mathbf{U}_R). \tag{9}$$

The grid motion velocity $\mathbf{w}_f^c$ at the center of the cell edge is usually set to be average of two node velocities

$$\mathbf{w}_f^c = \frac{1}{2}(\mathbf{w}_q + \mathbf{w}_{q^+}). \tag{10}$$

In these methods based on one-dimensional Riemann solver, the conservation of mass, momentum and total energy are kept naturally, and it's unnecessary to use index $c$ or $f$ to distinguish fluxes $\mathbf{F}_c^f$ and $\mathbf{F}_f^c$.

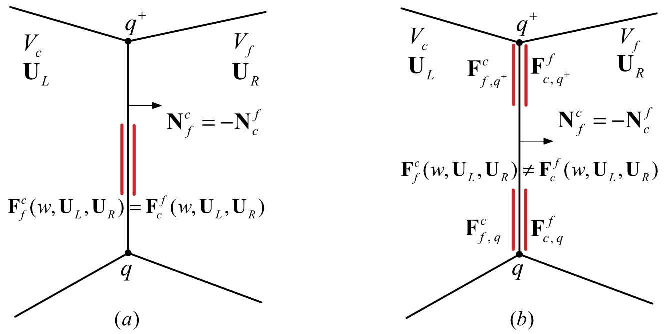


**Fig. 2.** Comparison between traditional ALE and new ALE methods. Fluxes across an edge of the cells can be different in the new method whereas they are the same in the methods based on one-dimensional Riemann solver.

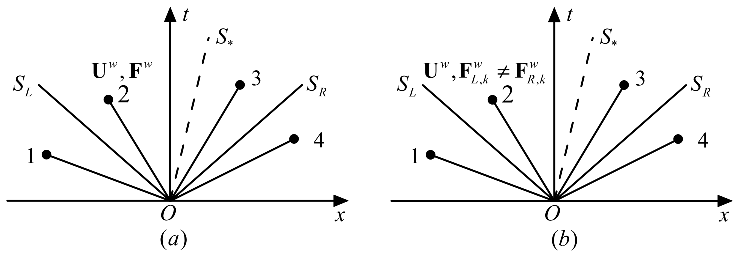


**Fig. 3.** Wave structures of the HLLC and HLLC-2D approximate Riemann solvers for the $x$-split two-dimensional equations. (a) HLLC, (b) HLLC-2D.

### 3.3. Two classical Riemann solvers

It is well known that a wide variety of Riemann solvers have been devised to solve the above Riemann problem [50]. In this paper we will consider Godunov and HLLC methods, which are shown in [34] to be stable in ALE formulation.

1) Godunov scheme on moving grids

Godunov method solves the exact Riemann fan and obtains all state values along each sampling direction $x/t$. The flux formula on moving grids is

$$\mathbf{F}^{c}_{f,Godunov}(w,\mathbf{U}_L,\mathbf{U}_R)=\mathbf{F}\big(\mathbf{U}^w\big)-w\mathbf{U}^w, \tag{11}$$

where $\mathbf{U}^w$ is the solution along $x/t=w$. The contact wave speed $S_*$ and pressure $p^*$ of the contact discontinuity are solved by an iterative method. For the convenience of the following discussion, the contact speed in a one-dimensional Riemann is also denoted by $\nu^*$. Now $\nu^*=S_*$.

2) HLLC scheme on moving grids

The HLLC approximate solver [49] includes 4 states and can accurately capture contact and shear waves for $x$-split two-dimensional gas dynamics, see Fig. 3(a). The vector of conservation quantities is

$$\mathbf{U}^w=\begin{cases}\mathbf{U}_L, & \text{if } w\leqslant S_L,\\ \mathbf{U}^*_L, & \text{if } S_L<w\leqslant S_*,\\ \mathbf{U}^*_R, & \text{if } S_*<w\leqslant S_R,\\ \mathbf{U}_R, & \text{if } w>S_R,\end{cases} \tag{12}$$

where $w=x/t$, and $\mathbf{U}^*_K$ for $K=L,R$ satisfy

$$\mathbf{U}^*_K=\rho_K\frac{S_K-u_K}{S_K-S_*}\begin{bmatrix}1\\ S_*\\ v_K\\ E_K+(S_*-u_K)(S_*+\frac{p_K}{\rho_K(S_K-u_K)})\end{bmatrix}, \tag{13}$$

where $S_L, S_R$ are the signal speeds of the left and right states. There are several choices for determining the sound speeds $S_R$ and $S_L$. A simple estimate is

$$S_L=\min(u_L-c_L,u_R-c_R),\qquad S_R=\max(u_L+c_L,u_R+c_R).$$

The intermediate speed $S_*$ in terms of $S_L$, $S_R$ is

$$S_* = v^* = \frac{p_L - p_R - (\rho u)_L(S_L - u_L) + (\rho u)_R(S_R - u_R)}{(\rho u)_L - (\rho u)_R + S_R\rho_R - S_L\rho_L}. \tag{14}$$

The corresponding HLLC flux function along $w = x/t$ is

$$\mathbf{F}^w = \begin{cases} \mathbf{F}_L, & \text{if } w \leqslant S_L, \\ \mathbf{F}_L^* = \mathbf{F}_L + S_L(\mathbf{U}_L^* - \mathbf{U}_L), & \text{if } S_L < w \leqslant S_*, \\ \mathbf{F}_R^* = \mathbf{F}_R - S_R(\mathbf{U}_R - \mathbf{U}_R^*), & \text{if } S_* < w \leqslant S_R, \\ \mathbf{F}_R, & \text{if } w \geqslant S_R. \end{cases} \tag{15}$$

In a framework of moving mesh, the numerical flux (6) of the HLLC solver can be represented as

$$\mathbf{F}^c_{f,HLLC}(w, \mathbf{U}_L, \mathbf{U}_R) = \begin{cases} \mathbf{U}_L(u_L - w) + \mathbf{D}_L, & \text{if } w \leqslant S_L, \\ \mathbf{U}_L^*(S_* - w) + \mathbf{D}^*, & \text{if } S_L < w \leqslant S_*, \\ \mathbf{U}_R^*(S_* - w) + \mathbf{D}^*, & \text{if } S_* < w \leqslant S_R, \\ \mathbf{U}_R(u_R - w) + \mathbf{D}_R, & \text{if } w > S_R, \end{cases} \tag{16}$$

where $G_K = (0, p, 0, pu)_K^T$ and $G^* = (0, p^*, 0, p^*S_*)^T$ express the force and work terms in different wave components. The contact pressure $p^*$ is

$$p^* = p_L + \rho_L(S_L - u_L)(S_* - u_L) = p_R + \rho_R(S_R - u_R)(S_* - u_R). \tag{17}$$

An equivalent HLLC flux formula is

$$\begin{aligned} \mathbf{F}^c_{f,HLLC}(w, \mathbf{U}_L, \mathbf{U}_R) = {} & \frac{1}{2}\big(\mathbf{F}(\mathbf{U}_L) - w\mathbf{U}_L + \mathbf{F}(\mathbf{U}_R) - w\mathbf{U}_R\big) \\ & - \frac{1}{2}\big(|S_L - w|\big(\mathbf{U}_L^* - \mathbf{U}_L\big) + |S_* - w|\big(\mathbf{U}_R^* - \mathbf{U}_L^*\big) + |S_R - w|\big(\mathbf{U}_R - \mathbf{U}_R^*\big)\big). \end{aligned} \tag{18}$$

We denote the ALE schemes (3) with the one-dimensional Godunov and HLLC Riemann solver as the ALE Godunov and ALE HLLC methods. Both the methods satisfy the consistency condition (8) and the conservation condition (9).

### 3.4. The CAVEAT algorithm

The ALE approach in CAVEAT code is a cell centered method, whose computation is performed in two phases: a Lagrangian phase, and a remapping phase in which conserved variables are transferred from a Lagrangian mesh to an arbitrary specified mesh, see [17] and [2]. For the sake of simplification, we focus only on its Lagrangian algorithm phase.

In Lagrangian frame, the control equations (2) reduce to

$$\begin{aligned} & \frac{d}{dt}\int_\Omega \rho \, dx\,dy = 0, \\ & \frac{d}{dt}\int_\Omega \rho\mathbf{u} \, dx\,dy + \int_{\partial\Omega} p\mathbf{N} \, dl = 0, \\ & \frac{d}{dt}\int_\Omega \rho E \, dx\,dy + \int_{\partial\Omega} p\mathbf{u}\cdot\mathbf{N} \, dl = 0. \end{aligned} \tag{19}$$

In addition, the variation in time of a control volume needs to satisfy the geometric conservation law (GCL)

$$\frac{d}{dt}\int_\Omega dx\,dy - \int_{\partial\Omega} \mathbf{u}\cdot\mathbf{N} \, dl = 0. \tag{20}$$

The CAVEAT approach is originally derived from the above equations. Suppose the discretization of the system of equations is built on an initial mesh as shown in Fig. 1(a), then the discrete scheme in CAVEAT code is similar to that in (3)

$$\begin{aligned} & \rho_c^{n+1} = \frac{|V_c^n|}{|V_c^{n+1}|}\rho_c^n, \\ & \rho_c^{n+1}\mathbf{u}_c^{n+1} = \frac{|V_c^n|}{|V_c^{n+1}|}\rho_c^n\mathbf{u}_c^n - \frac{\Delta t}{|V_c^{n+1}|}\sum_f L_f^c p_f^c \mathbf{N}_f^c, \\ & \rho_c^{n+1}E_c^{n+1} = \frac{|V_c^n|}{|V_c^{n+1}|}\rho_c^n E_c^n - \frac{\Delta t}{|V_c^{n+1}|}\sum_f L_f^c p_f^c v_f^c, \end{aligned} \tag{21}$$

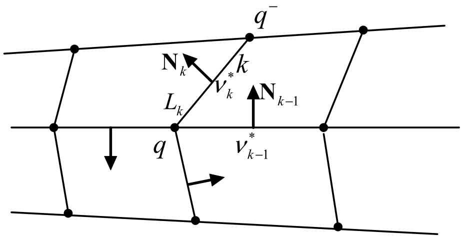


**Fig. 4.** Velocities associated with a cell vertex.

where the contact pressure and velocity $p_f^c$, $\nu_f^c$ are computed by solving an exactly or approximately one-dimensional Riemann problem in the normal direction of the edge $c \wedge f$.

To move the grid in a Lagrangian manner, the CAVEAT code provides a mesh evolution approach [17]. This algorithm requires that the projection of velocity at a vertex into the normal direction of an edge (sharing the vertex) should be equal to the contact velocity of a one-dimensional Riemann solution on that edge. Refer to Fig. 4 where the index $k$ is used to represent edge $c \wedge f$. Due to the fact that the problem is often overdetermined at a vertex, the vertex velocity $\mathbf{u}_q^*$ can be obtained by minimizing the following quadratic functional for each vertex $q$

$$Func(\mathbf{u}_q^*) = \sum_{k \in \mathcal{K}(q)} L_k \alpha_k (\mathbf{u}_q^* \cdot \mathbf{N}_k - \nu_k^*)^2, \tag{22}$$

where $\mathbf{N}_k$ is the unit normal direction of edge $k$ and $L_k$ is its length, and $\alpha_k$ is a weight defined on each edge $k$ (currently $\alpha_k = 1$ for these two Riemann solvers), and $\nu_k^*$ is the contact velocity $\nu_f^c$ along the norm direction of $k$.

The node position at the next time can be obtained by

$$\mathbf{x}_q^{n+1} = \mathbf{x}_q^n + \Delta t \mathbf{u}_q^*. \tag{23}$$

There exists two different discrete forms for the cell volume. One is consistent with flux form in (21),

$$|V_c^{n+1}| = |V_c^n| + \Delta t \sum_f L_f^c \nu_f^c. \tag{24}$$

Another is from the nodal coordinates $\mathbf{x}_q^{n+1}$, which is adopted by CAVEAT code and has the following approximate form

$$|V_c^{n+1}| = |V_c^n| + \frac{1}{2} \Delta t \sum_f L_f^c (\mathbf{u}_q^* + \mathbf{u}_{q+}^*) \cdot \mathbf{N}_f^c. \tag{25}$$

Making comparison (25) with (21), we can see that the flux computation in CAVEAT code is not compatible with the node displacement. To distinguish between the CAVEAT algorithm and the ALE one in Subsection 3.2, we denote the former (21) associated with the Godunov and HLLC solver as the Lagrangian Godunov and Lagrangian HLLC methods.

When the ALE scheme in Subsections 3.1–3.3 is used on a mesh whose node velocity is calculated via a least squares optimization (23), the advection term through cell edge usually does not vanish due to the difference between the edge moving velocity and the contact velocity.

In fact, the edge moving velocity $w$ in the edge flux (16), which is set to be the average of two node velocities (10)

$$w = 0.5(\mathbf{u}_q^* + \mathbf{u}_{q+}^*) \cdot \mathbf{N}_f^c,$$

in general, is not equal to the contact velocity, which is

$$S_* = \nu_f^c \neq w.$$

Consequently,

**Remark 1.** When the mesh moves via a least squares optimization, the CAVEAT approach, as a Lagrangian one, is not a natural reduction of the ALE approach (3) with (10), but it still can be understood as the formula (3) with a modified edge velocity $w = \nu_f^c$. Note that at this time the edge mesh velocity is incompatible with the nodal velocity.

**Remark 2.** The ALE method (3) with (10) has advantages in theory over the Lagrangian one (21): (1) the edge moving velocity in mass, momentum and energy fluxes keeps the consistent with nodal motion manner; (2) the 2-D unsteady Euler system of equations in certain ALE frameworks is hyperbolic whereas it degenerates to be weakly hyperbolic (lacking one eigenvector although all eigenvalues are real) in Lagrangian framework [27].

Unfortunately both of the ALE Godunov and ALE HLLC methods are still unstable under the interaction of the strong shock, and even more unstable than the Lagrangian one.

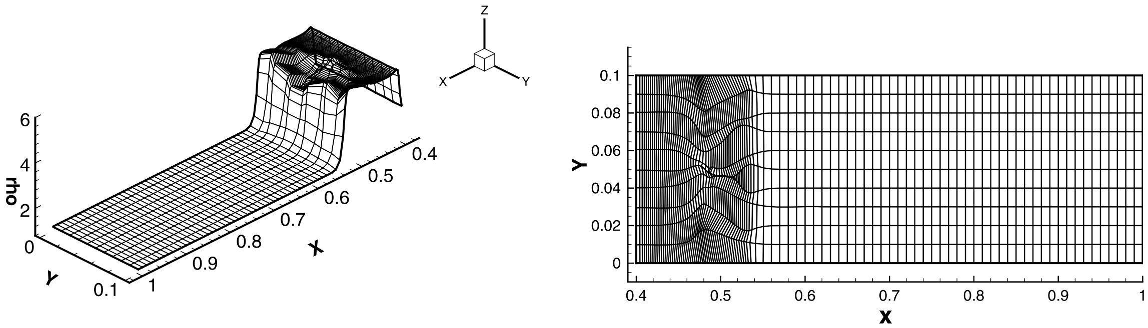


**Fig. 5.** Instability phenomena of the ALE Godunov method and distorted grid. Grid resolution is $100 \times 10$, and $e_0 = 10^{-8}$, $t = 0.4$.

## 4. An instability phenomenon with a numerical shock

### 4.1. Statement of problem

Let us describe the simplest case of a motion involving a shock front. The problem consists of a rectangular box whose walls form reflective boundaries and whose left-hand side wall acts as a piston, moving with constant velocity 1 towards the right into the (initially) static gas. The working fluid in front of shock is assumed to be an ideal gas with initial states

$$(\rho, u, v, p, \gamma) = \left(1, 0, 0, \frac{2e_0}{3}, \frac{5}{3}\right). \tag{26}$$

If the stationary gas is regarded as cold gas, i.e., $e_0 = 0$, then shock strength is infinity, otherwise it is finite.

The initial computational domain is $[0, 1] \times [0, 0.1]$, and the initial mesh has $M \times N$ uniform cells. The centerline of the grid is perturbed by

$$y_{i,jmid} = \begin{cases} y_{i,jmid} + 10^{-6}, & \text{if } i \text{ is even,} \\ y_{i,jmid} - 10^{-6}, & \text{if } i \text{ is odd.} \end{cases} \tag{27}$$

This mesh is called as "odd–even decoupling grid" in this paper which is often used to test the numerical shock instability in Eulerian calculation [44].

### 4.2. Instability phenomenon

In this section, we adopt sole mesh moving manner via the least square approach (22). Moreover, two solution strategies are used: one is the ALE method, whose mesh velocity (10) at the middle of the edge is different from the contact velocity derived from the Riemann solution, and the other is the Lagrangian CAVEAT method (21) in which no advection term appears.

When the two solution strategies are used in the piston problem, the numerical results display instability growth phenomena of the shock wave, which leads to a distorted grid. In the calculations, we let $e_0 = 10^{-8}$. Fig. 5 shows a density plot by using the ALE Godunov Riemann solver at $t = 0.4$. At that time the shock wave is far from the right wall (when $t = 0.75$ shock wave arrives right wall). The behaviors of numerical solutions of the ALE HLLC and ALE Godunov schemes are similar, so we illustrate only one of them in the following experiments for brevity.

By various numerical experiments the following two conclusions can be obtained.

(1) The shock strength is a key factor in the formation of the instability. When we increase the initial pressure to decrease the shock strength, the stability will be improved gradually. For example, when $e_0 = 0.1$, both of the two ALE methods can give satisfactory one-dimensional results till $t = 0.95$.

(2) The grid aspect ratio appears to be a significant factor in the forming of the instability phenomenon: very elongated grid cells that are normal to the shock can trigger instabilities, whereas meshes stretched along the tangent of the shock have a damping effect. Fig. 6 shows an unstable case for above example ($e_0 = 0.1$) when increasing mesh number in $y$ direction. Here initial grid resolution is $100 \times 20$.

Pakmor et al. [43] have noticed the instability of ALE methods for MHD problems and proposed a modification technique. They regard that the root of the instability is an error in the mass flux in the moving frame. In the circumstances of the grid velocity being close to the fluid velocity, the error in the mass flux can cause a sign change, destroying the upwind property of the scheme and making it unstable. They make a coordinate transformation to correct this behavior by calculating the fluxes in the fixed frame of the interface, then project the results to a moving frame. This method succeeds in MHD problems due to the effect of magnetic field, but it cannot be directly used to cure the instability arising in gas dynamics problems. In Appendix A, we describe Pakmor's method and point out that the transformed form is equivalent to the original Riemann solver for gas dynamics problems.

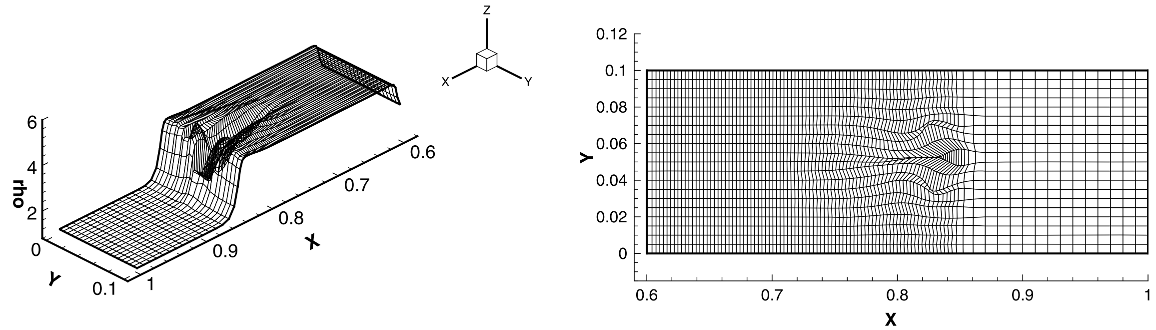


**Fig. 6.** Instability phenomena of the ALE Godunov method and distorted grid. Grid resolution is $100 \times 20$, and $e_0 = 0.1$, $t = 0.6$.

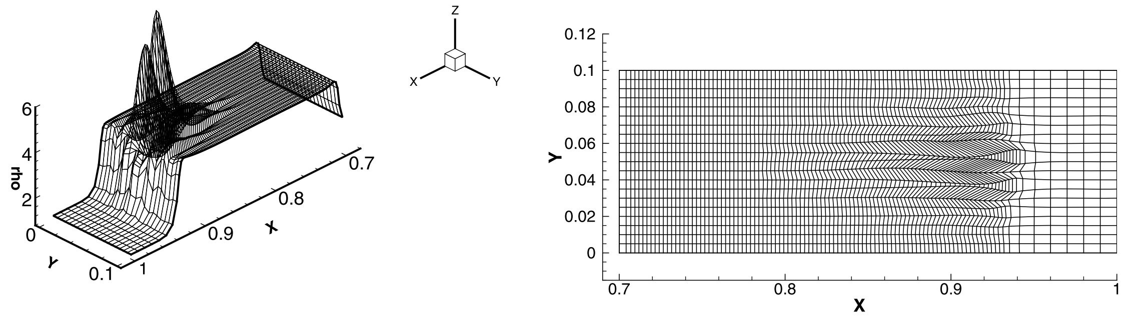


**Fig. 7.** Instability phenomena of the Lagrangian HLLC method and distorted grid. Grid resolution is $100 \times 20$, and $e_0 = 10^{-8}$, $t = 0.7$.

However, we know that adding negative dissipation via advection can cause a scheme to be unstable. In order to avoid the influence of advection errors in the mass flux, we use the Lagrangian methods (21) adopted in the CAVEAT code [17] and observe its behavior. The numerical experiments show that such modification indeed improves numerical stability. For the original example ($e_0 = 10^{-8}$), instability is restrained and calculation can continue to $t = 0.9$. But when we increase mesh number in $y$ direction, it comes forth again. See Fig. 7.

The numerical results for this difficult test problem clearly indicate one deficiency of the ALE HLLC and ALE Godunov schemes: serious instabilities under interaction of strong shocks. Furthermore, on a specific grid the Lagrangian flux can reduce the instability, but is difficult to eliminate it radically.

Dukowicz in [18] points out that the spurious vorticity error is one of the roots of an instability of the Lagrangian Godunov method (21). Maire et al. ascribe the instability to incompatibility of flux computation with node displacement and propose a robust Lagrangian method [36]. We agree with their viewpoints and will design a new ALE method. The object is to keep the edge fluxes consistency with the nodal flow, and furthermore to enhance the stability of the calculation in the framework of a moving mesh.

## 5. A new ALE method

The main feature of our new method is that the two-dimensional Riemann solver HLLC-2D in [46] is used. Two key ingredients are introduced in the construction of the method. One is a nodal solver, which is based on a vertex $q$ with multi-state initial values $\{\mathbf{U}_c, c \in \mathcal{C}(q)\}$ around the vertex $q$, which refer to Fig. 1(b). The nodal contact velocity $\mathbf{u}_q^*$ will be evaluated from this solver. Another one is the solver HLLC-2D, which is an extension of classical one-dimensional HLLC Riemann solver. The inputs of the Riemann problem are the nodal flow velocity $\mathbf{u}_q^*$, and the state vectors $\mathbf{U}_c$ and $\mathbf{U}_f$ between an edge $c \wedge f$ in Fig. 1(a). The purpose is to provide approximate flux formulae for $F_c^f$ and $F_f^c$.

### 5.1. Flux formula of HLLC-2D on moving mesh

In this subsection, we assume that the nodal contact velocity $u_q^*$ has been determined and focus on the solution of the one-dimensional Riemann solver whose initial states $(\mathbf{U}_c, \mathbf{U}_{c^+})$ are set between each edge on any vertex $q$.

To keep the consistency of description, let's return to Fig. 2. According to the direction that is normal to the cell edge, the initial values of the Riemann problem (5) between the edge are denoted by left and right data:

$$\mathbf{U}_L = \mathcal{T}_f^c \mathbf{U}_c, \qquad \mathbf{U}_R = \mathcal{T}_f^c \mathbf{U}_f.$$

The corresponding fluxes are $\mathbf{F}_L = \mathbf{F}(\mathbf{U}_L)$, $\mathbf{F}_R = \mathbf{F}(\mathbf{U}_R)$.

Different from traditional one-dimensional Riemann solver, we introduce an artificial contact wave speed $S_* = \mathbf{u}_q^* \cdot \mathbf{N}_f^c$ and construct an approximate Riemann solver HLLC-2D [46] for this Riemann problem.

Define two signal velocities $S_L$ and $S_R$ and let $S_L \leqslant S_* \leqslant S_R$. Therefore the HLLC-2D solver includes 4 state regions separated by $S_L$, $S_*$ and $S_R$. In each region, the vector of conservation quantities is defined as those in the HLLC solver:

$$\mathbf{U}^w = \begin{cases} \mathbf{U}_L, & \text{if } w \leqslant S_L, \\ \mathbf{U}_L^*, & \text{if } S_L < w \leqslant S_*, \\ \mathbf{U}_R^*, & \text{if } S_* < w \leqslant S_R, \\ \mathbf{U}_R, & \text{if } w > S_R, \end{cases} \tag{28}$$

where $\mathbf{U}_L^*$, $\mathbf{U}_R^*$ have the same formulae as (13). But flux functions are new, which are no longer the same as that in HLLC solver. In each state region two approximate fluxes are introduced:

$$\mathbf{F}_K^w = \begin{cases} \mathbf{F}_{K,1}, & \text{if } w \leqslant S_L, \\ \mathbf{F}_{K,2}, & \text{if } S_L < w \leqslant S_*, \\ \mathbf{F}_{K,3}, & \text{if } S_* < w \leqslant S_R, \\ \mathbf{F}_{K,4}, & \text{if } w > S_R, \end{cases} \tag{29}$$

where $K = L, R$, and they ensure that the jump conditions are satisfied strictly for each flux when the fluid flows through the discontinuous line between the different states. That is

$$\begin{aligned} &\mathbf{F}_{L,2} - \mathbf{F}_{L,1} = S_L\big(\mathbf{U}_L^* - \mathbf{U}_L\big) = \mathbf{F}_{R,2} - \mathbf{F}_{R,1}, \\ &\mathbf{F}_{L,3} - \mathbf{F}_{L,2} = S_*\big(\mathbf{U}_R^* - \mathbf{U}_L^*\big) = \mathbf{F}_{R,3} - \mathbf{F}_{R,2}, \\ &\mathbf{F}_{L,4} - \mathbf{F}_{L,3} = S_R\big(\mathbf{U}_R - \mathbf{U}_R^*\big) = \mathbf{F}_{R,4} - \mathbf{F}_{R,3}, \end{aligned}$$

where boundary fluxes are $\mathbf{F}_{L,1} = \mathbf{F}_L$, $\mathbf{F}_{R,4} = \mathbf{F}_R$ respectively.

The main reason of introducing two fluxes in each state region is that the artificial contact discontinuity velocity $S_*$ does not satisfy the equality (14), which is derived from one-dimensional Rankine–Hugoniot relations. For more details, refer to [46].

The numerical fluxes around a vertex $q$ on the moving mesh are

$$\mathbf{F}_{f,q}^c(w, \mathbf{U}_L, \mathbf{U}_R) = \mathbf{F}_L^w - w\mathbf{U}^w = \begin{cases} \mathbf{U}_L(u_L - w) + \mathbf{D}_L, & \text{if } w \leqslant S_L, \\ \mathbf{U}_L^*(S_* - w) + \mathbf{D}_L^*, & \text{if } S_L < w \leqslant S_*, \\ \mathbf{U}_R^*(S_* - w) + \mathbf{D}_L^*, & \text{if } S_* < w \leqslant S_R, \\ \mathbf{U}_R(u_R - w) + \mathbf{D}_R - \mathbf{D}_R^* + \mathbf{D}_L^*, & \text{if } w > S_R, \end{cases} \tag{30}$$

and

$$\mathbf{F}_{c,q}^f(w, \mathbf{U}_L, \mathbf{U}_R) = \mathbf{F}_R^w - w\mathbf{U}^w = \begin{cases} \mathbf{U}_L(u_L - w) + \mathbf{D}_L - \mathbf{D}_L^* + \mathbf{D}_R^*, & \text{if } w < S_L, \\ \mathbf{U}_L^*(S_* - w) + \mathbf{D}_R^*, & \text{if } S_L \leqslant w < S_*, \\ \mathbf{U}_R^*(S_* - w) + \mathbf{D}_R^*, & \text{if } S_* \leqslant w < S_R, \\ \mathbf{U}_R(u_R - w) + \mathbf{D}_R, & \text{if } w \geqslant S_R, \end{cases} \tag{31}$$

where $\mathbf{D}_K = (0, p, 0, pu)_K^T$, $\mathbf{D}_K^* = (0, p_K^*, 0, p_K^* S_*)^T$ for $K = L, R$ are vectors including the pressure and work terms on two ends and in star regions, respectively. Pressures are

$$\begin{cases} p_L^* = p_L + \rho_L(S_L - u_L)(S_* - u_L), \\ p_R^* = p_R + \rho_R(S_R - u_R)(S_* - u_R). \end{cases} \tag{32}$$

Notice that pressures are no longer necessarily continuous across the contact wave. This is a distinct feature of the HLLC-2D compared with one-dimensional Riemann solver. We need not worry about the pressure balance in the vicinity of the contact wave since a nodal conservation relation described in the following subsection can solve this balance problem.

If $S_*$ satisfies (14), then the Riemann solver HLLC-2D reduces to the classical HLLC solver. Such cases happen in some one-dimensional flows on some special meshes, or a uniform flow on a general mesh [46].

Referring to Fig. 2, a similar derivation can be done at vertex $q^+$, and flux $\mathbf{F}_{f,q^+}^c$ is available. For the whole edge flux there is

$$\mathbf{F}_f^c\big(w, \mathcal{T}_f^c\mathbf{U}_c, \mathcal{T}_f^c\mathbf{U}_f\big) = \frac{1}{2}\Big(\mathbf{F}_{f,q}^c\big(\mathbf{w}_q \cdot \mathbf{N}_f^c, \mathcal{T}_f^c\mathbf{U}_c, \mathcal{T}_f^c\mathbf{U}_f\big) + \mathbf{F}_{f,q^+}^c\big(\mathbf{w}_{q^+} \cdot \mathbf{N}_f^c, \mathcal{T}_f^c\mathbf{U}_c, \mathcal{T}_f^c\mathbf{U}_f\big)\Big). \tag{33}$$

When viewing the edge $c \wedge f$ from $V_f$, the local coordinate system changes and there is

$$\mathbf{N}_c^f = -\mathbf{N}_f^c, \qquad \bar{\mathbf{U}}_L = \mathcal{T}_c^f\mathbf{U}_f, \qquad \bar{\mathbf{U}}_R = \mathcal{T}_c^f\mathbf{U}_c.$$

Similar to Eqs. (30) and (31), we can obtain two numerical fluxes $\bar{\mathbf{F}}^f_{c,q}$ and $\bar{\mathbf{F}}^c_{f,q}$. The following claim shows that the fluxes $\bar{\mathbf{F}}^f_{c,q}$ and $\bar{\mathbf{F}}^c_{f,q}$ are consistent with $\mathbf{F}^f_{c,q}$ and $\mathbf{F}^c_{f,q}$ respectively.

**Claim 1.** *The fluxes $\bar{\mathbf{F}}^f_{c,q}$, $\bar{\mathbf{F}}^c_{f,q}$ and $\mathbf{F}^f_{c,q}$, $\mathbf{F}^c_{f,q}$ satisfy the following relations:*

$$\left(\mathcal{T}^f_c\right)^{-1}\bar{\mathbf{F}}^f_{c,q}\left(-w, \mathcal{T}^f_c\mathbf{U}_f, \mathcal{T}^f_c\mathbf{U}_c\right) = -\left(\mathcal{T}^c_f\right)^{-1}\mathbf{F}^f_{c,q}\left(w, \mathcal{T}^c_f\mathbf{U}_c, \mathcal{T}^c_f\mathbf{U}_f\right), \tag{34}$$

$$\left(\mathcal{T}^f_c\right)^{-1}\bar{\mathbf{F}}^c_{f,q}\left(-w, \mathcal{T}^f_c\mathbf{U}_f, \mathcal{T}^f_c\mathbf{U}_c\right) = -\left(\mathcal{T}^c_f\right)^{-1}\mathbf{F}^c_{f,q}\left(w, \mathcal{T}^c_f\mathbf{U}_c, \mathcal{T}^c_f\mathbf{U}_f\right). \tag{35}$$

**Proof.** Notice that $\bar{w} = \mathbf{w}_q \cdot \mathbf{N}^f_c = -\mathbf{w}_q \cdot \mathbf{N}^c_f = -w$ and from

$$\bar{\mathbf{U}}_L = \begin{bmatrix} \rho_R \\ -\rho_R u_R \\ -\rho_R v_R \\ \rho_R E_R \end{bmatrix}, \qquad \bar{\mathbf{U}}_R = \begin{bmatrix} \rho_L \\ -\rho_L u_L \\ -\rho_L v_L \\ \rho_L E_L \end{bmatrix},$$

there is the flux according to Eq. (30),

$$\bar{\mathbf{F}}^f_{c,q}(\bar{w}, \bar{\mathbf{U}}_L, \bar{\mathbf{U}}_R) = \begin{cases} \bar{\mathbf{U}}_L(\bar{u}_L - \bar{w}) + \bar{\mathbf{D}}_L, & \text{if } \bar{w} \leqslant \bar{S}_L, \\ \bar{\mathbf{U}}^*_L(\bar{S}_* - \bar{w}) + \bar{\mathbf{D}}^*_L, & \text{if } \bar{S}_L < \bar{w} \leqslant \bar{S}_*, \\ \bar{\mathbf{U}}^*_R(\bar{S}_* - \bar{w}) + \bar{\mathbf{D}}^*_L, & \text{if } \bar{S}_* < \bar{w} \leqslant \bar{S}_R, \\ \bar{\mathbf{U}}_R(\bar{u}_R - \bar{w}) + \bar{\mathbf{D}}_R - \bar{\mathbf{D}}^*_R + \bar{\mathbf{D}}^*_L, & \text{if } \bar{w} > \bar{S}_R. \end{cases}$$

The velocities and signal speeds in this local coordinate system are opposite to those with initial values $(\mathbf{U}_L, \mathbf{U}_R)$,

$$\bar{u}_L = -u_R, \qquad \bar{S}_L = -S_R, \qquad \bar{u}_R = -u_L, \qquad \bar{S}_R = -S_L, \qquad \bar{S}_* = -S_*.$$

Other states have the same signals,

$$\bar{\rho}^*_L = \rho^*_R, \qquad \bar{p}^*_L = p^*_R, \qquad \bar{E}^*_L = E^*_R, \qquad \bar{\rho}^*_R = \rho^*_L, \qquad \bar{p}^*_R = p^*_L, \qquad \bar{E}^*_R = E^*_L.$$

We can prove (34) for each situation of $w$. But without loss of generalization, we only consider the case $S_L < w \leqslant S_*$, i.e., $\bar{S}_* \leqslant \bar{w} < \bar{S}_R$. In the circumstance, the flux is

$$\bar{\mathbf{F}}^f_{c,q}(\bar{w}, \bar{\mathbf{U}}_L, \bar{\mathbf{U}}_R) = \bar{\mathbf{U}}^*_R(\bar{S}_* - \bar{w}) + \bar{\mathbf{G}}^*_L = \begin{bmatrix} \rho^*_L \\ -\rho^*_L S_* \\ -\rho^*_L v_L \\ \rho^*_L E^*_L \end{bmatrix}(-S_* + w) + \begin{bmatrix} 0 \\ p^*_R \\ 0 \\ -p^*_R S_* \end{bmatrix} = \begin{bmatrix} -\rho^*_L(S_* - w) \\ \rho^*_L S_*(S_* - w) + p^*_R \\ \rho^*_L v_L(S_* - w) \\ -(\rho^*_L E^*_L(S_* - w) + p^*_R S_*) \end{bmatrix}.$$

Note that in case $S_L < w \leqslant S_*$ the flux (31) has the following form:

$$\mathbf{F}^f_{c,q}(w, \mathbf{U}_L, \mathbf{U}_R) = \mathbf{U}^*_L(S_* - w) + \mathbf{D}^*_R = \begin{bmatrix} \rho^*_L \\ \rho^*_L S_* \\ \rho^*_L v_L \\ \rho^*_L E^*_L \end{bmatrix}(S_* - w) + \begin{bmatrix} 0 \\ p^*_R \\ 0 \\ p^*_R S_* \end{bmatrix} = \begin{bmatrix} \rho^*_L(S_* - w) \\ \rho^*_L S_*(S_* - w) + p^*_R \\ \rho^*_L v_L(S_* - w) \\ \rho^*_L E^*_L(S_* - w) + p^*_R S_* \end{bmatrix}.$$

Comparing the above equality, there holds

$$\left(\mathcal{T}^f_c\right)^{-1}\bar{\mathbf{F}}^f_{c,q}(\bar{w}, \bar{\mathbf{U}}_L, \bar{\mathbf{U}}_R) = -\left(\mathcal{T}^c_f\right)^{-1}\mathbf{F}^f_{c,q}(w, \mathbf{U}_L, \mathbf{U}_R).$$

Likewise,

$$\left(\mathcal{T}^f_c\right)^{-1}\bar{\mathbf{F}}^c_{f,q}(\bar{w}, \bar{\mathbf{U}}_L, \bar{\mathbf{U}}_R) = -\left(\mathcal{T}^c_f\right)^{-1}\mathbf{F}^c_{f,q}(w, \mathbf{U}_L, \mathbf{U}_R).$$

We finish the proof of Claim 1. □

Due to the consistency between $\bar{\mathbf{F}}^f_{c,q}$ and $\mathbf{F}^f_{c,q}$, we can drop off the overbar '−' of $\bar{\mathbf{F}}^f_{c,q}$ and do not lead to any confusion.

Following the same methodology, we know Claim 1 also holds for the half edge fluxes connected to the node $q^+$. Furthermore, according to the definition of the flux (33), the consistency condition (8) on the whole edge is satisfied.

**Claim 2.** *There is*

$$\mathbf{F}^c_{f,q}(w,\mathbf{U}_L,\mathbf{U}_R)-\mathbf{F}^f_{c,q}(w,\mathbf{U}_L,\mathbf{U}_R)=\mathbf{D}^*_L-\mathbf{D}^*_R. \tag{36}$$

In summary, the new method is of two characteristics being different from traditional one:

(1) Introduce a unique contact velocity $\mathbf{u}^*_q$ at each grid vertex $q$.
(2) Introduce different fluxes $\mathbf{F}^c_{f,q}$, $\mathbf{F}^c_{f,q^+}$, on each edge of $V_c$. Edge flux $\mathbf{F}^c_f$ viewed from $V_c$ is no longer equal to $\mathbf{F}^f_c$ necessarily, which is viewed from $V_f$, due to the nodal contact velocity introduced.

We will evaluate the $\mathbf{u}^*_q$ in a consistent way to ensure that the scheme keeps locally conservation of mass, momentum and total energy.

### 5.2. Conservation relations

Since $\mathbf{F}^c_f\neq\mathbf{F}^f_c$ may happen, the conservation across edges is no longer retained, and the conservation of mass, momentum and total energy need be enforced at nodes. We deduce this procedure from the global conservation in domain $\Omega$.

Referring to Fig. 1 and making summation to Eq. (3) over all cells, there are

$$\begin{aligned}
\frac{d}{dt}\int\limits_{\Omega}\mathbf{U}\,dx\,dy &\approx \frac{1}{\Delta t}\sum_c\left(\left|V_c^{n+1}\right|\mathbf{U}_c^{n+1}-\left|V_c^n\right|\mathbf{U}_c^n\right)\\
&=-\sum_c\sum_{f\in\mathcal{F}(c)}L^c_f\left(\mathcal{T}^c_f\right)^{-1}\mathbf{F}^c_f\left(\mathbf{w}\cdot\mathbf{N}^c_f,\mathcal{T}^c_f\mathbf{U}_c,\mathcal{T}^c_f\mathbf{U}_f\right)\\
&=-\frac{1}{2}\sum_q\sum_{c\in\mathcal{C}(q)}\left[L^c_{c^+}\left(\mathcal{T}^c_{c^+}\right)^{-1}\mathbf{F}^c_{c^+,q}\left(\mathbf{w}_q\cdot\mathbf{N}^c_{c^+},\mathcal{T}^c_{c^+}\mathbf{U}_c,\mathcal{T}^c_{c^+}\mathbf{U}_{c^+}\right)\right.\\
&\quad\left.+L^c_{c^-}\left(\mathcal{T}^c_{c^-}\right)^{-1}\mathbf{F}^c_{c^-,q}\left(\mathbf{w}_q\cdot\mathbf{N}^c_{c^-},\mathcal{T}^c_{c^-}\mathbf{U}_c,\mathcal{T}^c_{c^-}\mathbf{U}_{c^-}\right)\right]\\
&\equiv:-\frac{1}{2}\left(\sum_{q\in\mathcal{I}(\Omega)}\sum_{c\in\mathcal{C}(q)}\mathbf{FF}_{c,q}+\sum_{q\in\mathcal{B}(\Omega)}\sum_{c\in\mathcal{C}(q)}\mathbf{FF}_{c,q}\right),
\end{aligned}\tag{37}$$

where $\mathcal{B}(\Omega)$ is the set of boundary nodes, and $\mathcal{I}(\Omega)$ represents the set of interior nodes. Here we have replaced the global summation over cells by a summation over nodes. $\mathbf{FF}_{c,q}$ are the outward normal fluxes cross the edges of cell $V_c$ connected to the vertex $q$, see Fig. 1(b).

The discrete conservation law requires

$$\sum_{q\in\mathcal{I}(\Omega)}\sum_{c\in\mathcal{C}(q)}\mathbf{FF}_{c,q}=0.$$

A sufficient condition to retain the conservation is

$$\sum_{c\in\mathcal{C}(q)}\mathbf{FF}_{c,q}=0,\quad \text{if } q\in\mathcal{I}(\Omega). \tag{38}$$

Notice that

$$\begin{aligned}
\sum_{c\in\mathcal{C}(q)}\mathbf{FF}_{c,q}&=\sum_{c\in\mathcal{C}(q)}L^c_{c^+}\left[\left(\mathcal{T}^c_{c^+}\right)^{-1}\mathbf{F}^c_{c^+,q}\left(\mathbf{w}_q\cdot\mathbf{N}^c_{c^+},\mathcal{T}^c_{c^+}\mathbf{U}_c,\mathcal{T}^c_{c^+}\mathbf{U}_{c^+}\right)+\left(\mathcal{T}^{c^+}_c\right)^{-1}\mathbf{F}^{c^+}_{c,q}\left(\mathbf{w}_q\cdot\mathbf{N}^{c^+}_c,\mathcal{T}^{c^+}_c\mathbf{U}_{c^+},\mathcal{T}^{c^+}_c\mathbf{U}_c\right)\right]\\
&=\sum_{c\in\mathcal{C}(q)}L^c_{c^+}\left(\mathcal{T}^c_{c^+}\right)^{-1}\left[\mathbf{F}^c_{c^+,q}\left(\mathbf{w}_q\cdot\mathbf{N}^c_{c^+},\mathcal{T}^c_{c^+}\mathbf{U}_c,\mathcal{T}^c_{c^+}\mathbf{U}_{c^+}\right)-\mathbf{F}^{c^+}_{c,q}\left(\mathbf{w}_q\cdot\mathbf{N}^c_{c^+},\mathcal{T}^c_{c^+}\mathbf{U}_c,\mathcal{T}^c_{c^+}\mathbf{U}_{c^+}\right)\right],
\end{aligned}\tag{39}$$

where a shift from $c^-\wedge c$ to $c\wedge c^+$ is performed in the first equality and the consistency relation (8) is used in the second equality.

Referring to Fig. 1(b) again, we use $k$ to represent the edge $c\wedge c^+$. $\mathbf{U}_c$ and $\mathbf{U}_{c^+}$ are left and right states of edge $k$ respectively. By use of (36) in Claim 2, there is

$$\sum_{c\in\mathcal{C}(q)}\mathbf{FF}_{c,q}=\sum_{c\in\mathcal{C}(q)}L^c_{c^+}\left(\mathcal{T}^c_{c^+}\right)^{-1}\left(\mathbf{D}^*_{L,k}-\mathbf{D}^*_{R,k}\right)=\sum_{k\in\mathcal{K}(q)}L_k\left(p^*_{L,k}-p^*_{R,k}\right)(\mathcal{T}_k)^{-1}(0,1,0,S_{*,k})^T=0, \tag{40}$$

where $\mathcal{K}(q)$ is the set of edges sharing the vertex $q$. It follows that

$$\sum_{k\in\mathcal{K}(q)} L_k\big(p^*_{L,k}-p^*_{R,k}\big)\mathbf{N}_k=0, \tag{41}$$

and from $S_{*,k}=\mathbf{N}_k\cdot\mathbf{u}^*_q$, there holds

$$\sum_{k\in\mathcal{K}(q)} L_k\big(p^*_{L,k}-p^*_{R,k}\big)\mathbf{N}_k\cdot\mathbf{u}^*_q=0. \tag{42}$$

The condition (41) implies the condition (42). Therefore once (41) holds, the local nodal conservation including mass, momentum and total energy will be preserved.

### 5.3. Nodal solver

After substituting (32) into (41), we obtain

$$\sum_{k\in\mathcal{K}(q)} L_k(\alpha_{L,k}+\alpha_{R,k})\big[\mathbf{u}^*_q\cdot\mathbf{N}_k-\nu^*_k\big]\mathbf{N}_k=0, \tag{43}$$

where

$$\alpha_{L,k}=-\rho_{L,k}(S_{L,k}-u_{L,k}),\qquad \alpha_{R,k}=\rho_{R,k}(S_{R,k}-u_{R,k}), \tag{44}$$

and $\nu^*_k$ is the contact velocity in the classical one-dimensional HLLC Riemann solver (14) for edge $k$

$$\nu^*_k=\frac{p_{L,k}-p_{R,k}+\alpha_{L,k}u_{L,k}+\alpha_{R,k}u_{R,k}}{\alpha_{L,k}+\alpha_{R,k}}. \tag{45}$$

We solve Eq. (43) to get the contact velocity $\mathbf{u}^*_q$ at each vertex $q$. Then we get

$$\mathbf{u}^*_q=M^{-1}\sum_{k\in\mathcal{K}(q)} L_k(\alpha_{L,k}+\alpha_{R,k})\nu^*_k\mathbf{N}_k, \tag{46}$$

where

$$M=\begin{bmatrix}\sum_{k\in\mathcal{K}(q)} L_k(\alpha_{L,k}+\alpha_{R,k})n^2_{x,k} & \sum_{k\in\mathcal{K}(q)} L_k(\alpha_{L,k}+\alpha_{R,k})n_{x,k}n_{y,k}\\ \sum_{k\in\mathcal{K}(q)} L_k(\alpha_{L,k}+\alpha_{R,k})n_{x,k}n_{y,k} & \sum_{k\in\mathcal{K}(q)} L_k(\alpha_{L,k}+\alpha_{R,k})n^2_{y,k}\end{bmatrix}.$$

**Remark 3.** The idea of nodal Riemann solver in the above two subsections has been used extensively in the context of Lagrangian methods, e.g., see [11,16,36,40]. Though the nodal contact velocity in this paper is derived from the conservation of mass, momentum and total energy in ALE form, the resulting formula is the same as that by Maire et al. in [36].

From now on we denote the ALE scheme (3) with the flux (33) as the ALE HLLC-2D method, in which two fluxes $\mathbf{F}^c_{f,q}$ and $\mathbf{F}^c_{f,q^+}$ in (30) and the contact velocity at the vertex (46) are used.

**Remark 4.** When grid moving velocity $\mathbf{w}_q$ is equal to $\mathbf{u}^*_q$ in (46), the advection terms $\mathbf{U}_K(S_*-w)$ in the flux formula (30) vanish naturally. We denote the ALE HLLC-2D as the Lagrangian HLLC-2D at the time. This is in contrast with the ALE methods using one-dimensional Riemann solver in Section 3, see Remark 1. In this regard, the ALE HLLC-2D provides a unified representation from Eulerian to Lagrangian method only according to different grid moving velocities.

Just as Maire et al. pointed out in [36], the Lagrangian velocity (46) has another interpretation. The left hand side of Eq. (43) is the gradient of the following quadratic functional,

$$Func\big(\mathbf{u}^*_q\big)=\sum_{k\in\mathcal{K}(q)}(\alpha_{L,k}+\alpha_{R,k})\big(\mathbf{u}^*_q\cdot\mathbf{N}_k-\nu^*_k\big)^2. \tag{47}$$

Therefore the contact velocity in (46) is equivalent to the solution from the minimization of the functional (47). Notice that the function (47) differs from CAVEAT function (22) in that an arbitrary weight parameter $\alpha_k$ in (22) at the edge $k$ is replaced by a specific $\alpha_{L,k}+\alpha_{R,k}$.

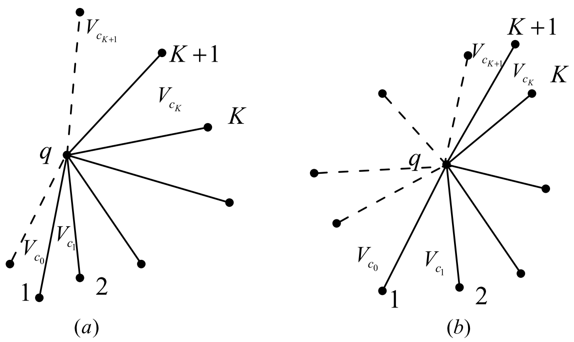


**Fig. 8.** Notations for the boundary conditions: (a) boundary nodes are not collinear, (b) boundary nodes are collinear.

*5.4. Boundary conditions*

In the ALE formalism, we adopt the classical zero-order extrapolation method to implement boundary conditions. For velocity boundaries such as solid wall or piston, we need to distinguish two cases. (1) When three boundary nodes are not collinear, we implement simple extrapolation like that in Fig. 8(a). In this case, only two ghost cells are introduced. (2) When three boundary nodes are collinear, we implement mirror extrapolation, i.e., a symmetrical ghost point is introduced for each vertex connecting a boundary node, referring to Fig. 8(b). Such a method provides the same formula (46) as that for internal vertex when calculating $\mathbf{u}_q^*$.

*5.5. Time step limitation*

We distinguish two different moving mesh strategies. For Eulerian method with fixed grid, we use the classical CFL criterion, otherwise we use the Lagrangian time step limitation proposed by Maire et al. [36].

At time $t^n$, for each cell $V_c$ we denote by $\lambda_c$ the in-circle diameter of the cell. The classical CFL criterion is

$$\Delta t_E = C_E \min_c \frac{\lambda_c}{\sqrt{u_c^2 + v_c^2} + c_c}, \tag{48}$$

where $C_E$ is a strictly positive coefficient and $c_c$ is the sound speed in the cell. Usually $C_E = 0.3$.

Maire et al. suggest the time step to be adopted

$$\Delta t = \min(\Delta t_E, \Delta t_V, C_M \Delta t^n), \tag{49}$$

where $\Delta t^n$ is the current time step and $C_M$ is a coefficient which is set to be $C_M = 1.01$. $\Delta t_V$ limits the variation of volume

$$\Delta t_V = C_V \min_c \{ |V_c^n| / |V_c'| \},$$

where $C_V = 0.1$ and

$$|V_c'| = \frac{1}{2} \sum_f L_f^c (\mathbf{u}_q^* + \mathbf{u}_{q^+}^*) \cdot \mathbf{N}_f^c.$$

*5.6. Description of the algorithm procedure*

**Step 1.** Initialization

At time $t = t^n$ we know all flow variables in each cell: $\rho_c^n$, $\mathbf{u}_c^n$, $p_c^n$, $e_c^n$, $c_c^n$ and the geometrical quantities: $\mathbf{x}_q^n$, $V_c^n$, $\mathbf{N}_f^c$, $L_f^c$.

**Step 2.** Nodal solver

The contact velocity $\mathbf{u}_q^*$ at each vertex is obtained by solving the linear system of equations in (46).

**Step 3.** Time step limitation

The time step limitation $\Delta t$ is obtained from (48) or (49) for Eulerian or moving mesh method.

**Step 4.** Grid moving strategy

In the ALE method, specific grid moving strategy, such as harmonic mapping equation (B.1) or interpolation technique, needs to be used to determine vertex coordinate $\mathbf{x}_q^{n+1}$ at $t^{n+1}$. In Eulerian methods, the vertex coordinate $\mathbf{x}_q^{n+1} = \mathbf{x}_q^n$ and in Lagrangian methods $\mathbf{x}_q^{n+1} = \mathbf{x}_q^n + \Delta t \mathbf{u}_q^*$. Hence the nodal motion velocity $\mathbf{w}_q = (\mathbf{x}_q^{n+1} - \mathbf{x}_q^n)/\Delta t$. Furthermore, the cell volume $V_c^{n+1}$ are obtained from the vertex coordinates.

**Step 5.** The Riemann solver HLLC-2D

At the common edge $c \wedge f$ between the cell $V_c$ and $V_f$, the half edge fluxes $\mathbf{F}_{f,q}^c$ and $\mathbf{F}_{f,q^+}^c$ are calculated from (30). The edge flux $\mathbf{F}_f^c$ is the average of the two half edge fluxes (33).

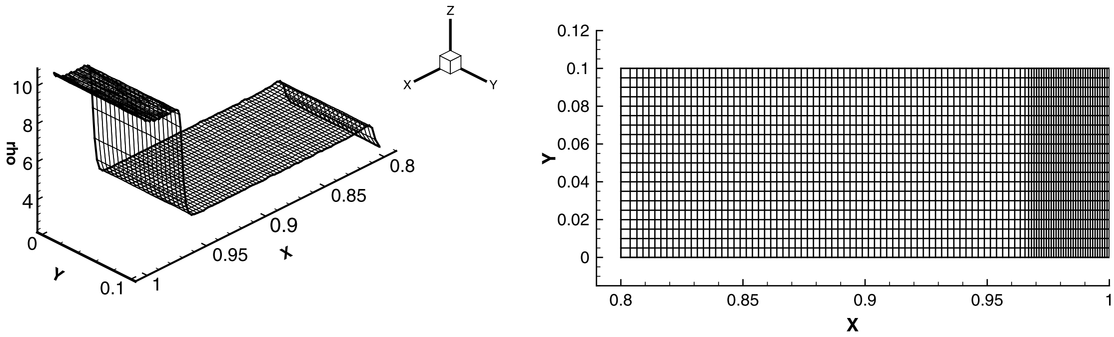


**Fig. 9.** The density and grid of the ALE HLLC-2D method. Grid resolution is $100 \times 20$, and $e_0 = 10^{-8}$, $t = 0.8$.

**Step 6.** Update
The conservation quantities in cell $V_c$ are updated by (3). It goes to Step 1 if the calculation continues, otherwise it stops.

If the second order accurate method is adopted, then an MUSCL type reconstruction for physical quantities is performed. The flow variables in Step 1 are replaced by those reconstructed ones in the vicinity of the vertex. Refer to [46] for more details.

## 6. Numerical experiments

In the following experiments, four kinds of grids are used to illustrate the robustness and performance of the proposed method.

The first grid is denoted as 'the grid A'. The grid moving velocity $\mathbf{w}_q$ is equal to $\mathbf{u}_q^*$ in (46) or (47), on which the ALE HLLC-2D method degenerates to the Lagrangian HLLC-2D. We will present some examples to stress on the ability of the Lagrangian HLLC-2D to restrain nonphysical vorticity errors.

The second is denoted as 'the grid B'. The nodal mesh velocity $\mathbf{w}_q$ is available from the function (22), but we purposely let weigh parameter $\alpha_k$ is not equal to $\alpha_{L,k} + \alpha_{R,k}$ in (47), for example, $\alpha_k = 1$. Due to the discrepancy between the node mesh velocity $\mathbf{w}_q$ and the contact velocity $\mathbf{u}_q^*$ in (47), the advection terms always arise in the numerical flux of the ALE HLLC-2D method, and these terms form a persistent disturbance to the Lagrangian HLLC-2D method. We need such mesh velocity $\mathbf{w}_q$ to evaluate the robustness of the ALE HLLC-2D method to resist instability.

The third grid ('the grid C') is suitable for quadrilateral one, which is formed by a uniform linear interpolation of the region boundary grid nodes. The motion of boundary grid is determined by the least square method (22).

The last moving mesh strategy is an adaptive unstructured grid proposed by Chen et al. [14], which needs to solve an elliptic equation (B.1). A brief description of the algorithm is given in Appendix B. It needs to point out that all initial triangular meshes are generated by a Delaunay triangulation algorithm, and the mesh size is taken to be the maximum diameter of the circumcircles of the triangles in the domain.

In what follows, all test examples adopt CFL number $C_E = 0.25$.

### 6.1. A piston problem

In this subsection, we will compute the piston motion problem (26) in Section 4, but with different initial grids and grid moving strategies. Here we only consider the $e_0 = 10^{-8}$, which represents a strong shock case.

#### 6.1.1. Odd–even decoupling grid

The first example includes those tests in Section 4 using the ALE HLLC-2D method. The initial mesh is (27) and the grid strategy is 'the grid B'. Here we let the grid resolution is $100 \times 20$ which corresponds to the most unstable case. Since the initial grid disturbance in those examples is very small, the ALE HLLC-2D method keeps one-dimensional shock results rather well till $t = 0.8$. After that time, a slight two-dimensional perturbation will appear, but the calculation can continue to $t = 0.98$. Fig. 9 presents the numerical result at $t = 0.8$, we may compare it with Fig. 7 and know that the new ALE method is more robust than those based on the one-dimensional Riemann solver.

#### 6.1.2. Saltzmann problem

The Saltzmann test problem [18] is usually used to verify the robustness of a numerical scheme when the mesh is not aligned with the fluid flow. The initial mesh is generated by following formulae

$$x_{i,j} = (i-1)h_x + \big[0.1 - (j-1)h_y\big]\sin\big(\pi(i-1)h_x\big),$$
$$y_{i,j} = (j-1)h_y,$$

where $h_x = h_y = 0.01$, $1 \leqslant i \leqslant 101$, $1 \leqslant j \leqslant 11$.

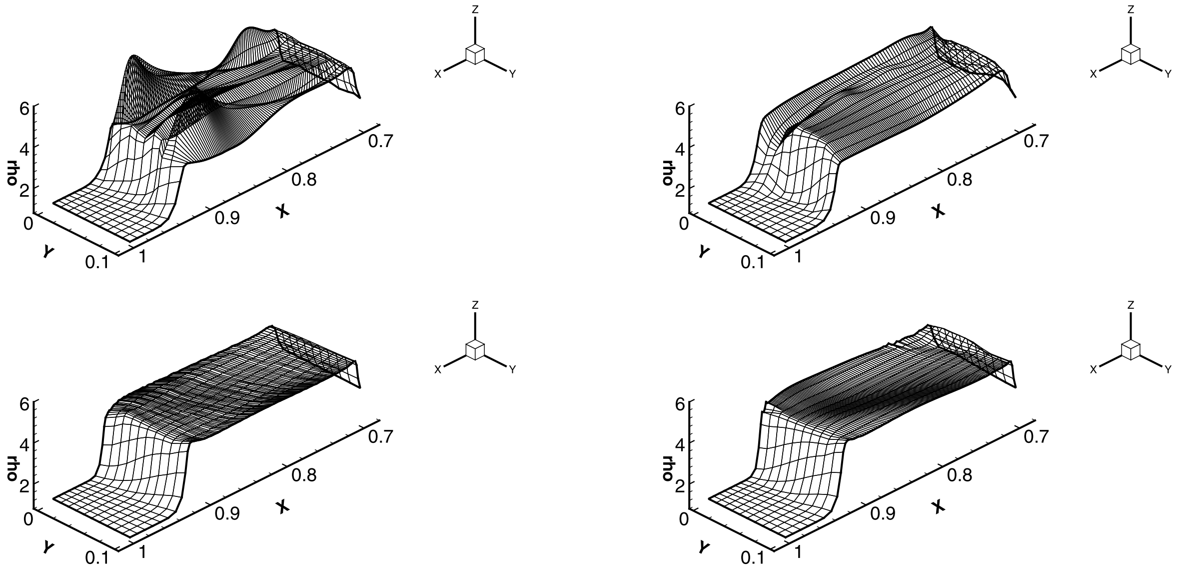


**Fig. 10.** Densities of Saltzmann problem using different methods with two different grid strategies at $t = 0.7$. Left: the grid B; right: the grid C; upper: the Lagrangian HLLC and ALE HLLC; bottom: the ALE HLLC-2D.

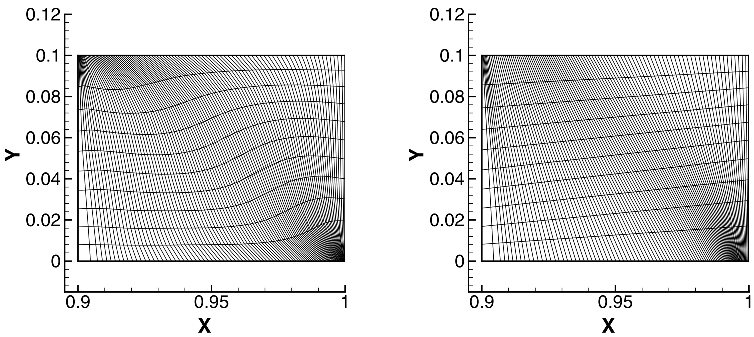


**Fig. 11.** Two different grids of Saltzmann problem using the ALE HLLC-2D method at $t = 0.9$. Left: the grid B, right: the grid C.

We adopt two different grid moving methods: (1) the grid B, (2) the grid C.

Due to grid tangling and collapse of the calculation in early stage when using the ALE HLLC method with the grid B, we use the Lagrangian HLLC method instead of ALE HLLC method to ensure calculation continue till $t = 0.75$. The Lagrangian HLLC method differs from the ALE HLLC in that the advection terms vanish and is more stable than the latter. For the grid C, the ALE HLLC can reach time $t = 0.75$ and then the calculation will stop. However the ALE HLLC-2D can overrun $t = 0.91$ for the two grids. Fig. 10 shows the densities at $t = 0.7$ obtained by using the different methods. Fig. 11 illustrates the two grids using the ALE HLLC-2D at $t = 0.9$.

*6.1.3. Hourglass grid problem*

The hourglass phenomenon usually appears in the Lagrangian calculation as a kind of unstable mode, see [35] and [12]. It arises because the discrete equations do not necessarily have a force to resist the hourglass-type motion. The grid is often tangled during the computation.

A possible solution to this problem is to introduce subzonal quantities, such as subzonal forces [12] or subzonal entropy [15]. These methods stabilize Lagrangian numerical scheme quite well, but they introduce some assumptions simultaneously as well, which are not always physical.

We consider the following initial grid,

$$x_{i,j} = (i-1)h_x + (-1)^{i+j}\delta,$$
$$y_{i,j} = (j-1)h_y,$$

where $h_x = h_y = 0.01$, $\delta = h_x/8$, $1 < i < 101$, $1 < j < 11$.

The strategy we move the mesh is ‘the grid A’, and the ALE HLLC-2D method reduces to the Lagrangian one naturally. The first order Lagrangian HLLC-2D scheme still tangles the grid, but computations with the second order scheme improve

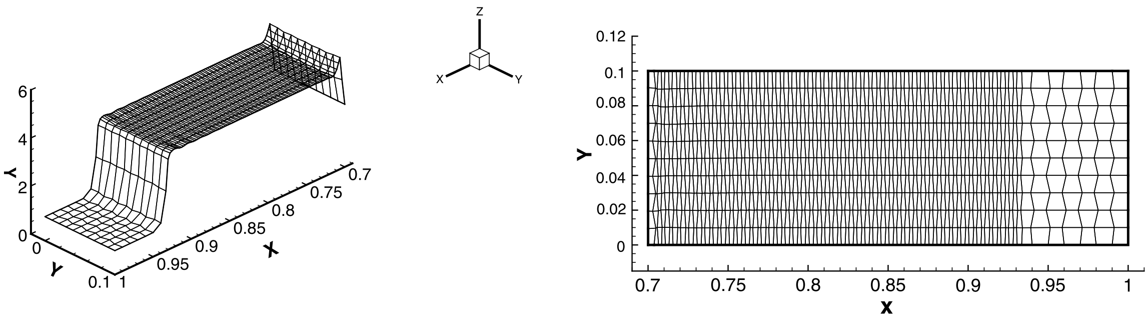


**Fig. 12.** A piston problem with an initial hourglass mesh using the Lagrangian HLLC-2D method at $t = 0.7$. Left: density, right: grid.

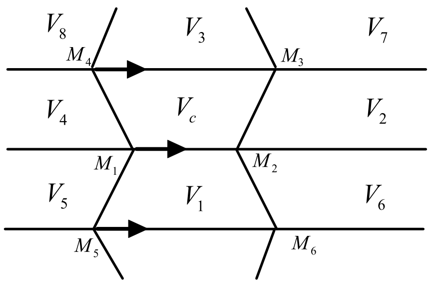


**Fig. 13.** A sketch of the above hourglass problem. When the vertex $M_1$ is very close to $M_2$, the grid distortion will appear. The second order approach may modify the nodal velocities at $M_1$, $M_4$, $M_5$.

the solution significantly. Fig. 12 illustrates density and grid at $t = 0.7$. Except for "wall heat" errors in the vicinity of piston, the hourglass grids are corrected rather well. This calculation is able to reach time $t = 0.95$ which corresponds to multiple, successive rebounds of the shock wave.

By Fig. 13 we explain the phenomenon heuristically. In this one-dimensional flow, the contact velocities at vertices $M_1$, $M_4$ can be proved to keep the same when using the first order accurate scheme. This is because the one-dimensional Riemann velocities around $M_4$ are always the same as ones around $M_1$ respectively. When $M_1$ is very close to $M_2$, the grid distortion will appear. Just like pointed in [46], the second order reconstruction of the physical quantities, they are implemented on nodes rather than cell edges, can change the distributions around $M_1$ and $M_4$ and thus can reduce the motion velocity of $M_1$ and increase velocity of $M_4$.

### 6.2. Shock refraction problem

This problem is proposed by Dukowicz and Meltz [18] which is also a piston driven problem. The significance of this problem is that it presents both physical vorticity and spurious grid distortion [18]. The computational domain is composed of two regions. At the initial time, the states and regions are

$$\begin{aligned}
&\text{Region 1:}\quad \rho_1 = 1,\ u_1 = 0,\ v_1 = 0,\ p_1 = 1,\ \gamma_1 = 1.4,\\
&\text{coordinate of vertexes:}\quad (0,0),\ (1,0),\ (1 + 1.5\sqrt{3}, 1.5),\ (0, 1.5)\\
&\text{Region 2:}\quad \rho_2 = 1.5,\ u_2 = 0,\ v_2 = 0,\ p_2 = 1,\ \gamma_2 = 1.4,\\
&\text{coordinate of vertexes:}\quad (1,0),\ (3,0),\ (3 + 1.5\sqrt{3}, 1.5),\ (1 + 1.5\sqrt{3}, 1.5).
\end{aligned}$$

The initial grid is formed by a uniform linear interpolation of the region boundary grid nodes, see Fig. 14. The cell resolutions are $36 \times 30$ and $40 \times 30$ in each region. In this problem a piston with the velocity 1.48 moves from the left sending a shock wave through an initially stationary gas in region 1. This shock then becomes incident condition on the second grid. Therefore transmitted shock, reflective shock and a shear motion along contact line are generated. In the calculation reflective boundary conditions are applied to the top and bottom of both grids. Fig. 15 shows the grid at time $t = 1.3$ just before the shock starts to run off the second grid. At the same time, an enlarged portion of the grid in the lower left hand corner next to the driving piston and lower boundary is displayed. We see that no grid tangling occurs in this region compared with some traditional methods [12]. The density contour in Fig. 16 shows a sharp contact discontinuity at the initial boundary of the two regions. All grid tangling has disappeared although horizontal meshes passing through contact line have large distortion due to the presence of a slip line.

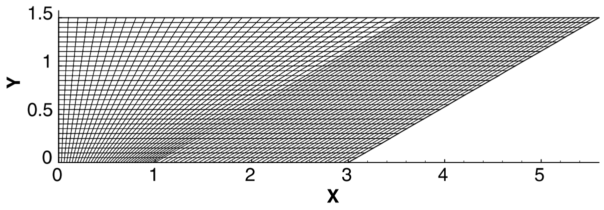


**Fig. 14.** Initial grid for Dukowicz's shock refraction problem.

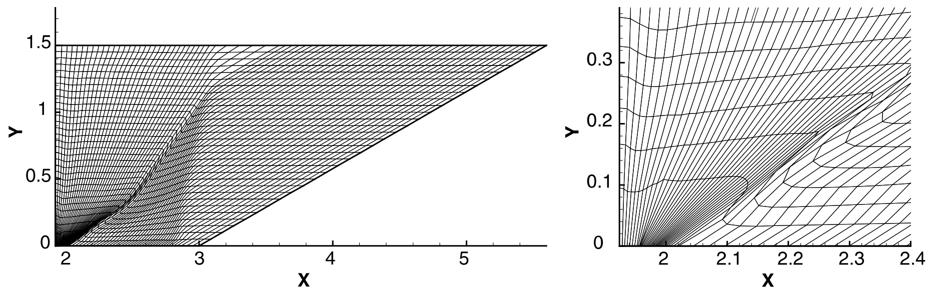


**Fig. 15.** Grid of the Dukowicz shock refraction problem at $t = 1.3$. Right is a zoom near left and lower corner.

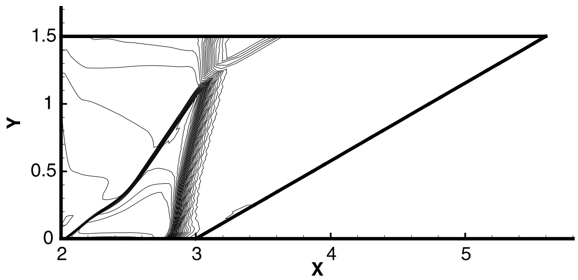


**Fig. 16.** Density contour of the Dukowicz shock refraction problem at $t = 1.3$. Thirty equally spaced contour lines from $\rho = 1.1$ to $\rho = 4.4$.

### 6.3. Steady vortex

We consider a 2-D inviscid vortex proposed by Yee et al. [55], for which the entropy is uniform. The vortex is initially located at the origin. Its velocity components $u$, $v$ and the absolute temperature $T$ are defined in non-dimensional form as:

$$u = u_0 - \frac{\Gamma}{2\pi} y \exp\left(\frac{1-r^2}{2}\right), \qquad v = v_0 - \frac{\Gamma}{2\pi} x \exp\left(\frac{1-r^2}{2}\right),$$
$$T = 1 - \frac{(\gamma-1)\Gamma^2}{8\gamma\pi^2} \exp\left(1-r^2\right), \tag{50}$$

where $r^2 = x^2 + y^2$, the $(u_0, v_0)$ is the convection velocity. The vortex strength $\Gamma$ is set equal to 5 and the ratio specific heats is $\gamma = 1.4$. In order to carry out the numerical convergence studies for both the Eulerian and Lagrangian approaches, we let $(u_0, v_0) = (0, 0)$.

The square-shaped computational domain is $\Omega = [-5, 5] \times [-5, 5]$. Periodic boundary conditions are applied in both directions. We know that the observed evolution is due uniquely to the numerical errors, therefore the test is a good example to check the accuracy of a numerical scheme. However, in a Lagrangian scheme, the mesh motion follows the fluid flow and quickly the grid becomes twisted, so that it is not possible to run this test problem for large times. Fig. 17 shows a Lagrangian mesh configuration and some density contours at $t = 5$ (20 equally spaced contour lines from $\rho = 0.6$ to $\rho = 0.96$). Compared with the Eulerian method, the numerical errors of the Lagrangian method become serious at the time. In order to measure the convergent rates of the Eulerian and Lagrangian scheme, we run this test case on a series of successively refined meshes and the corresponding error is expressed in the continuous $L_2$ norm as

$$Err(L_2) = \sqrt{\sum_c \left(\rho_c - \bar{\rho}(\mathbf{x}_c)\right)^2 |V_c|},$$

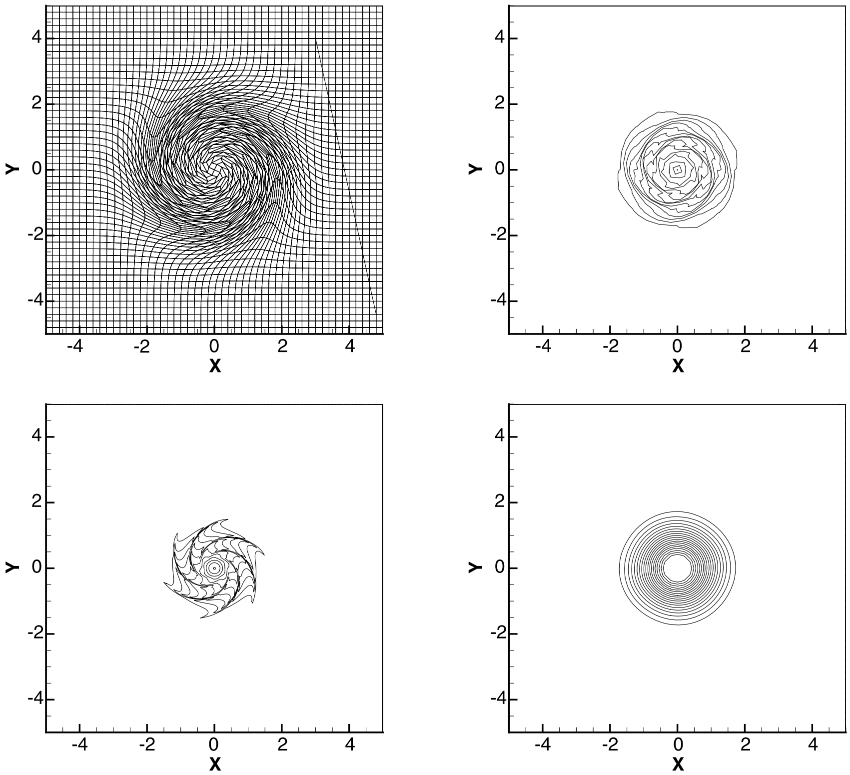


**Fig. 17.** Mesh configuration and the density contours for the isentropic vortex using the HLLC-2D approach at $t = 5$. Right and top: Lagrangian and quadangular mesh, left and bottom: Lagrangian and triangular mesh, right and bottom: Eulerian and quadangular mesh.

**Table 1**
Numerical convergence results of the Eulerian and Lagrangian algorithms using the HLLC-2D solver on quadrateral meshes.

| Cells | Euler | | Lagrange | |
|---|---|---|---|---|
| | $Err(L_2)$ | Order | $Err(L_2)$ | Order |
| 1st order | | | | |
| $25 \times 25$ | 2.39376e−01 | – | 1.97338e−01 | – |
| $50 \times 50$ | 1.53446e−01 | 0.64 | 1.33248e−01 | 0.57 |
| $100 \times 100$ | 8.76673e−02 | 0.81 | 7.96005e−02 | 0.74 |
| $200 \times 200$ | 4.70773e−02 | 0.90 | 4.39244e−02 | 0.86 |
| 2nd order | | | | |
| $25 \times 25$ | 6.86974e−02 | – | 9.790965e−02 | – |
| $50 \times 50$ | 1.51191e−02 | 2.18 | 2.146051e−02 | 2.19 |
| $100 \times 100$ | 2.99994e−03 | 2.33 | 3.560933e−03 | 2.59 |
| $200 \times 200$ | 6.30926e−04 | 2.25 | 8.590342e−04 | 2.05 |

where $\bar{\rho}(\mathbf{x}_c)$ is the exact solution of the density. The initial spatial steplengthes of four meshes are 0.4, 0.2, 0.1 and 0.05 respectively, where the steplength of a triangle mesh is taken to be the maximum diameter of the circumcircles of the triangles, refer to [8]. The final time is taken $t = 1$. Tables 1 and 2 show the resulting numerical convergence rates for the Eulerian and Lagrangian formulations on quadrangular and triangular meshes. The order of accuracy is well preserved for the first and second order methods on quadrangular meshes. The convergent rates on triangle meshes are good enough for the Eulerian method and the first order Lagrangian approach, but only attain first order accuracy for the second order Lagrangian method. The concrete reasons are not clear, and how to increase the accuracy of the Lagrangian approach on triangular mesh depends on a further research in the future.

### *6.4. Single material Sod shock tube problem*

The test is devoted to comparing different results using the Eulerian, Lagrangian and ALE HLLC-2D methods for 1D Sod shock tube problem computed on a 2D unstructured mesh. The computational domain is $[-0.5, 0.5] \times [0, 0.1]$ with an initial triangulation of 201 vertices in the $x$-direction and 21 vertices in the $y$-direction (2538 cells). Initial values are

**Table 2**
Numerical convergence results of the Eulerian and Lagrangian algorithms using the HLLC-2D solver on triangular meshes.

| Cells | Euler | | Lagrange | |
|---|---|---|---|---|
| | $Err(L_2)$ | Order | $Err(L_2)$ | Order |
| 1st order | | | | |
| 1490 | 2.41106e−01 | – | 2.45237e−01 | – |
| 5680 | 1.55101e−01 | 0.64 | 1.67429e−01 | 0.55 |
| 23 330 | 8.98663e−02 | 0.79 | 1.00513e−01 | 0.74 |
| 92 716 | 4.87077e−02 | 0.88 | 5.55814e−02 | 0.85 |
| 2nd order | | | | |
| 1490 | 7.86962e−02 | – | 2.671459e−01 | – |
| 5680 | 2.38539e−02 | 1.72 | 1.376016e−01 | 0.96 |
| 23 330 | 7.38849e−03 | 1.69 | 6.872518e−02 | 1.00 |
| 92 716 | 2.27191e−03 | 1.70 | 3.431974e−02 | 1.00 |

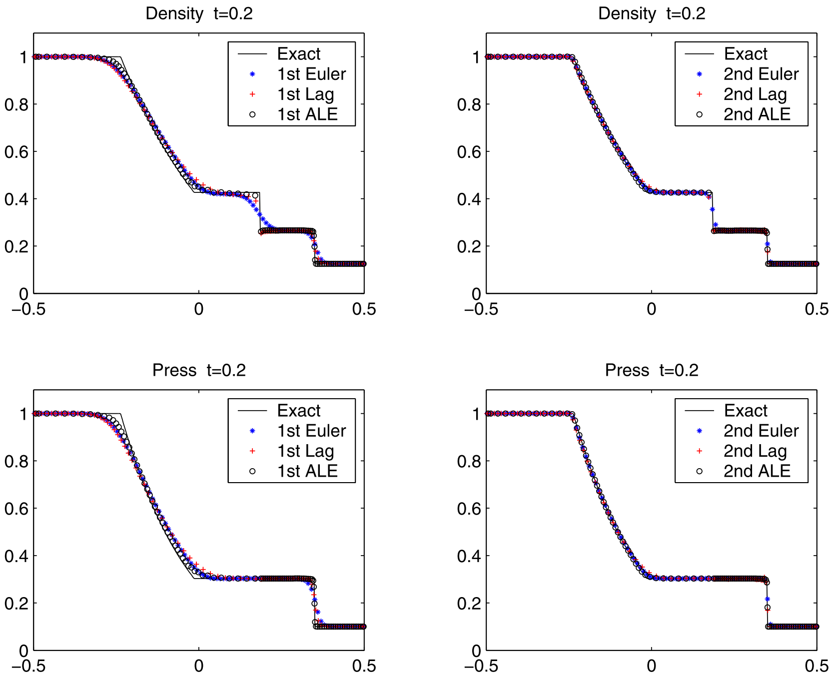


**Fig. 18.** Comparisons of density and pressure for the Eulerian, Lagrangian and ALE HLLC-2D methods in the Sod problem.

$$\begin{cases} (\rho_L, u_L, v_L, p_L, \gamma_L) = (1, 0, 0, 1, 1.4), & \text{if } -0.5 \leqslant x < 0, \\ (\rho_R, u_R, v_R, p_R, \gamma_R) = (0.125, 0, 0, 0.1, 1.4), & \text{if } 0 \leqslant x < 0.5. \end{cases}$$

The ALE method uses the adaptive moving mesh technique [14], which is described in Appendix B for completeness and some parameters in the monitor function are taken as follows: $\beta = 0.1$, $\alpha_\rho = 0$ and $\alpha_p = 50$. A new feature in this grid generation method is that we keep the contact discontinuity explicitly tracked as a Lagrangian curve.

Fig. 18 shows the one-dimensional density and pressure on a section plane parallel $x$-axis for three HLLC-2D schemes. Both first- and second-order accurate numerical approximations are compared to the exact solution. Fig. 19 illustrates grids of the Eulerian and ALE HLLC-2D methods. Fig. 20 displays the density and pressure from every zone in the entire mesh.

We can see that the first order accurate Eulerian method has more dissipation in the vicinity of contact discontinuity and the Lagrangian method has lower accuracy near the rarefaction wave. However our ALE method improves the solution's resolution for both contact and rarefaction waves.

### 6.5. Shock tube problem with different materials

This problem was proposed by Abgrall for calculating multicomponent flow [1]. The initial values are

$$\begin{cases} (\rho_L, u_L, v_L, p_L, \gamma_L) = \left(14.54903, 0, 0, 2.9 \times 10^7, 1.67\right), & \text{if } -0.5 \leqslant x < 0, \\ (\rho_R, u_R, v_R, p_R, \gamma_R) = \left(1.16355, 0, 0, 2.5 \times 10^5, 1.4\right), & \text{if } 0 \leqslant x < 0.5. \end{cases}$$

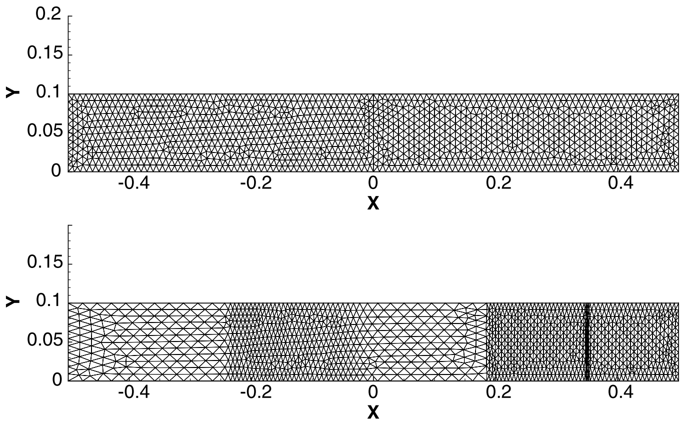


**Fig. 19.** Grids for the Sod problem using the Eulerian and ALE HLLC-2D methods, $t = 0.2$.

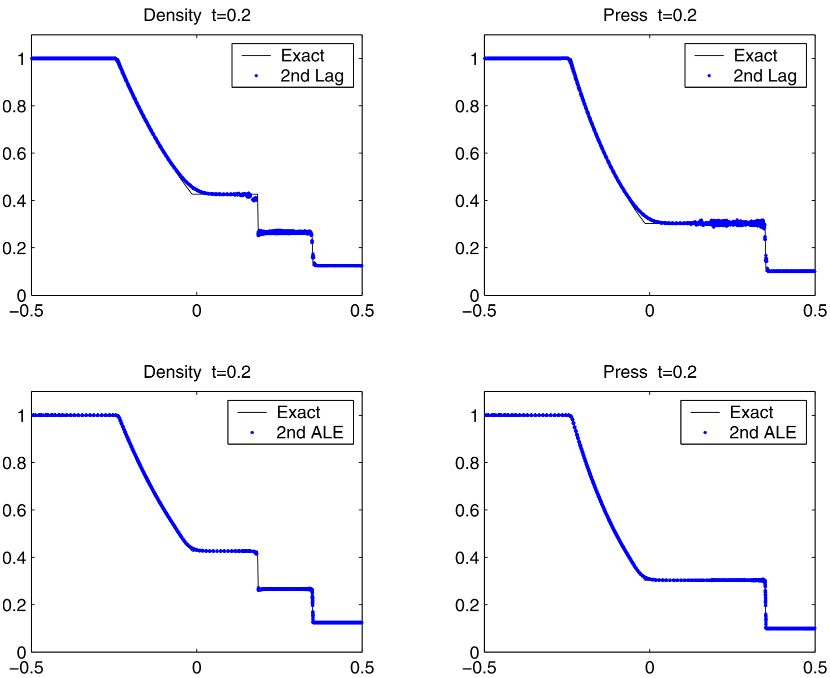


**Fig. 20.** Comparisons of density and pressure for the second order Lagrangian and ALE HLLC-2D methods in the Sod problem. Every cell center value in the mesh is plotted.

Because of the difficulty in preventing the numerical oscillation of the pressure and velocity fields, this Riemann problem is very interesting and is attracting many researchers. We calculate this problem on a 2D mesh and the computational domain is $[-0.5, 0.5] \times [0, 0.1]$. To avoid pressure oscillation near the material interface, we use the Lagrangian and ALE HLLC-2D methods whose material interface is tracked explicitly.

We only consider an unstructured mesh with the same initial grid as that in previous example.

Fig. 21 compares the density and pressure at $t = 0.0002$ using the second order accurate Lagrangian and ALE HLLC-2D schemes. Every cell center value in the mesh is plotted in this figure. Fig. 22 presents two grids given by the Lagrangian and ALE HLLC-2D methods respectively at that time. Both grids have large grid aspect near material interface. This phenomenon is caused by the strong rarefaction to uniform initial grid distribution. However, mesh distribution of ALE method at final time is more reasonable than Lagrangian one because the former is well-proportioned in compression and expansion region. Due to a strong rarefaction wave and wall heating errors in this problem, the Lagrangian method has oscillations in the rear of the shock wave. There are few oscillations in the ALE HLLC-2D calculation on the adaptive mesh method.

This example clearly demonstrates the potential of the ALE HLLC-2D scheme for solving multimaterial flows with strong rarefaction wave and contact discontinuity.

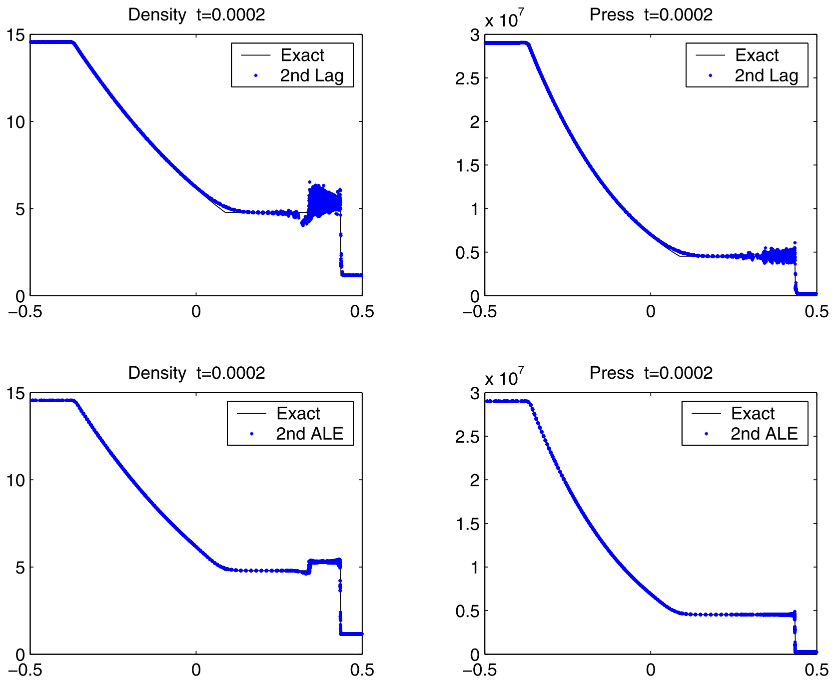


**Fig. 21.** Comparison between the Lagrangian and ALE HLLC-2D methods in Abgrall's problem. Every cell center value in the mesh is plotted.

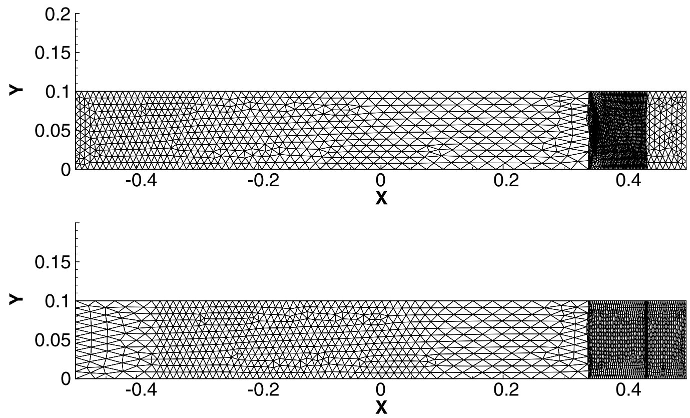


**Fig. 22.** Grids of the second order Lagrangian HLLC-2D (up) and ALE HLLC-2D (down) methods in Abgrall's problem, $t = 0.0002$.

### 6.6. Kinked Mach stem problem

This problem was studied by Woodward and Colella [53]. We use it to illustrate accuracy and robustness of adaptive mesh method. Initially, a Mach 10 shock is located at $x = -0.1$ and travels in the $x$-direction and encounters a wedge that makes an angle 30° with the shock propagation direction. The states in the undisturbed air in front of the shock are

$$\rho = 1.4, \qquad u = 0, \qquad p = 1, \qquad \gamma = 1.4.$$

The vertex coordinates of computational domain are

$$[-0.1, 0], \quad [0, 0], \quad [2.4, 2.4/\sqrt{3}], \quad [2.4, 2.1], \quad [-0.1, 2.1]$$

and computations run up to the dimensionless time $t = 0.2$. In such case, a double-Mach reflection (DMR) is formed. For some schemes in Eulerian methods, the primary Mach stem is sometimes kinked, similar to the carbuncle phenomenon. In [46], we point out that the HLLC-2D solver is stable even if using a very fine grid. Here we use a coarse grid. This quasi-uniform triangulation is formed by 5417 nodes, 15 912 edges, and 10 495 cells. This example takes $\alpha_\rho = 50$, $\alpha_p = 50$, $\beta = 0.1$. Fig. 23 shows density contours and the grid plot. The ALE HLLC-2D is able to offer accurate and robust solutions for capturing strong shock on arbitrarily moving grids.

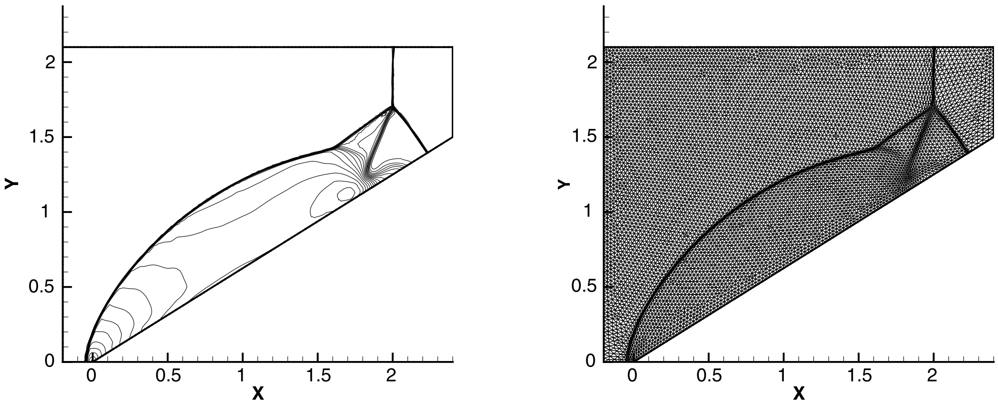


**Fig. 23.** Density contour and grid of Kinked Mach stem problem using the ALE HLLC-2D method at $t = 0.2$.

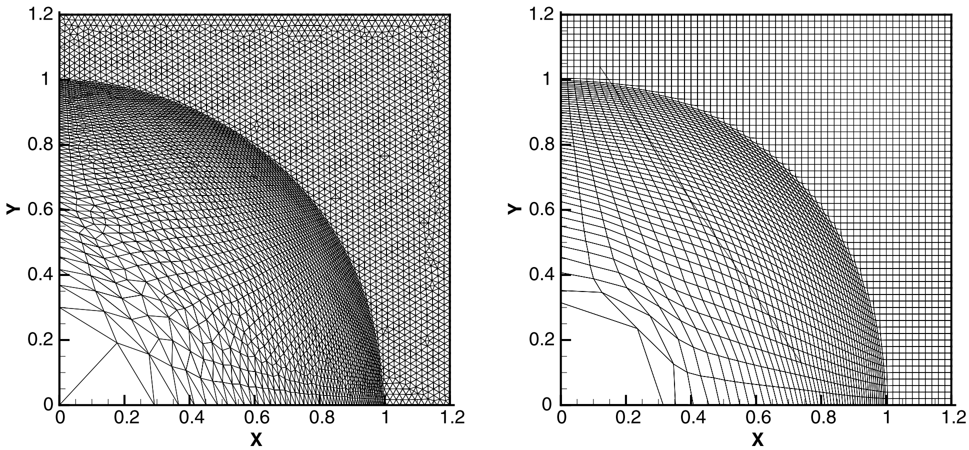


**Fig. 24.** The grid using the second order accurate Lagrangian HLLC-2D method in the Sedov problem at time $t = 1$. Left: the triangular mesh, right: the quadrilateral mesh.

### 6.7. Sedov problem

Sedov [45] obtained an analytic solution for a blast wave that is generated by a point source. We model this problem on a domain $[0, 1.2] \times [0, 1.2]$ with both quadrilateral and triangular meshes. The quadrilateral mesh adopts 60 cells in each direction, and the Delaunay triangular mesh has a similar mesh size, whose cell number is 8760. The initial state is

$$(\rho, p, u, v, \gamma) = \left(1, 10^{-6}, 0, 0, 5/3\right).$$

A finite source internal energy at the origin is $e_0 = 0.564113$. It will produce a shock that is located at radius 1 at time $t = 1$. The specific internal energy near the origin is initialized with 1 cell for the quadrilateral mesh and two cells for the triangular mesh. The values are $e_0/(4|V_c|)$, where $|V_c|$ is the sum of the cell volumes connected to the origin, and the factor 4 represents our calculation is on the first quadrant.

In this test example, we use the Lagrangian HLLC-2D method. Since both the 1st order and 2nd order approaches give similar grids, we only show the 2nd order results. Fig. 24 shows the grids at time $t = 1$ on the quadrilateral and triangular meshes. As seen here, the shock is traveling radially outward. The density and pressure fields on the two meshes are plotted in Figs. 25 and 26 respectively. Notice that physical quantities in each cells appear in the latter two figures to exhibit the symmetry of the numerical results. Clearly, both the first order and second order accurate results on the quadrilateral mesh are well symmetric, and the 2nd order approach has better accuracy and give a good agreement with the analytic solutions. Although the triangular mesh is not symmetric, the results still keep the symmetry rather well. Due to the triangular grid has more cell numbers than quadrilateral one in this test, the first order approach has enough accuracy, but the second order method has better symmetry than that of the first order approach on the triangular mesh.

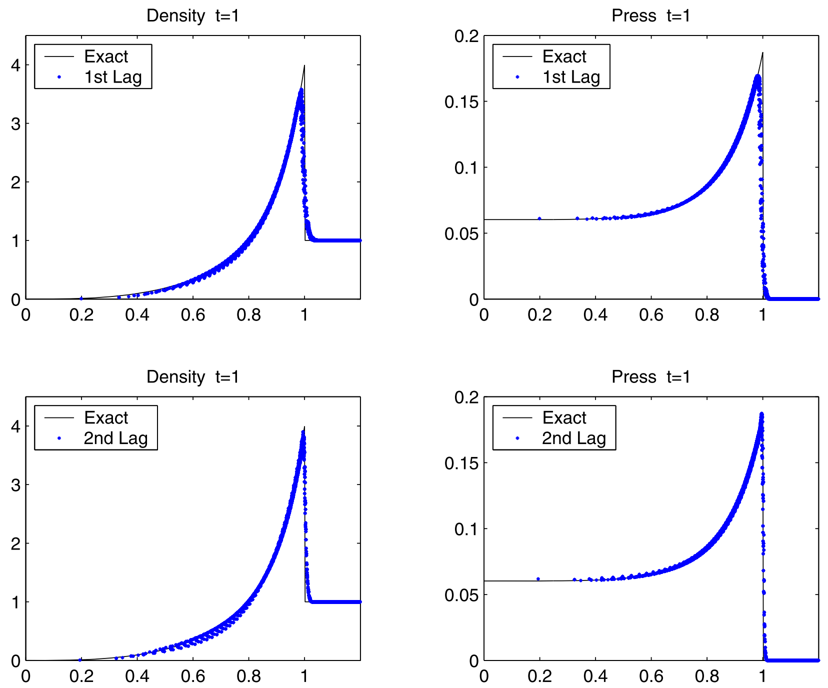


**Fig. 25.** The Sedov results using a quadrilateral mesh at $t = 1$. Top: the 1st order solution, bottom: the 2nd order solution. Every cell center value in the mesh is plotted.

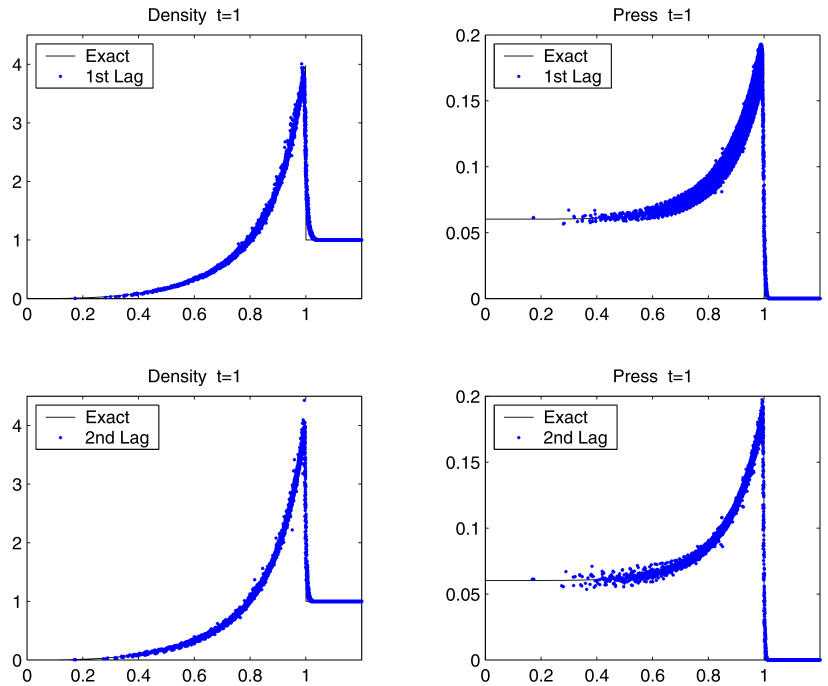


**Fig. 26.** The Sedov results using a triangular mesh at $t = 1$. Top: the 1st order solution, bottom: the 2nd order solution. Every cell center value in the mesh is plotted.

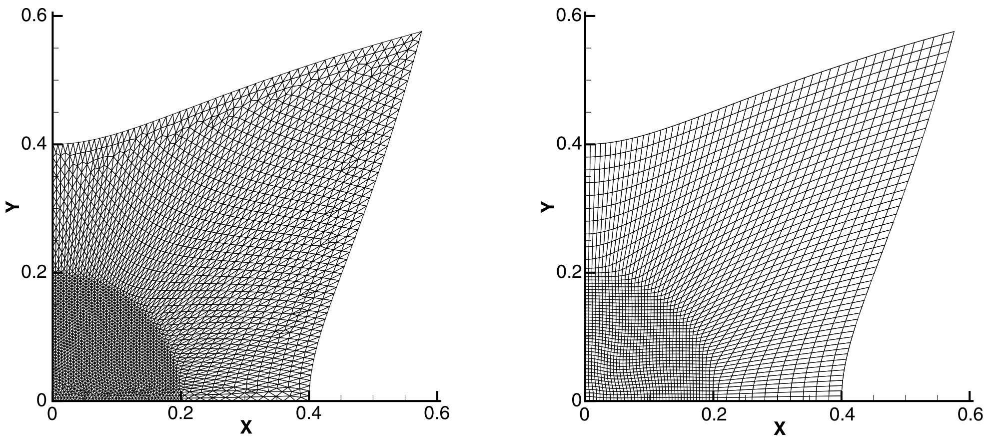


**Fig. 27.** The grids using the Lagrangian HLLC-2D approach in the Noh problem at time $t = 0.6$. Left: the triangular mesh, right: the quadrilateral mesh.

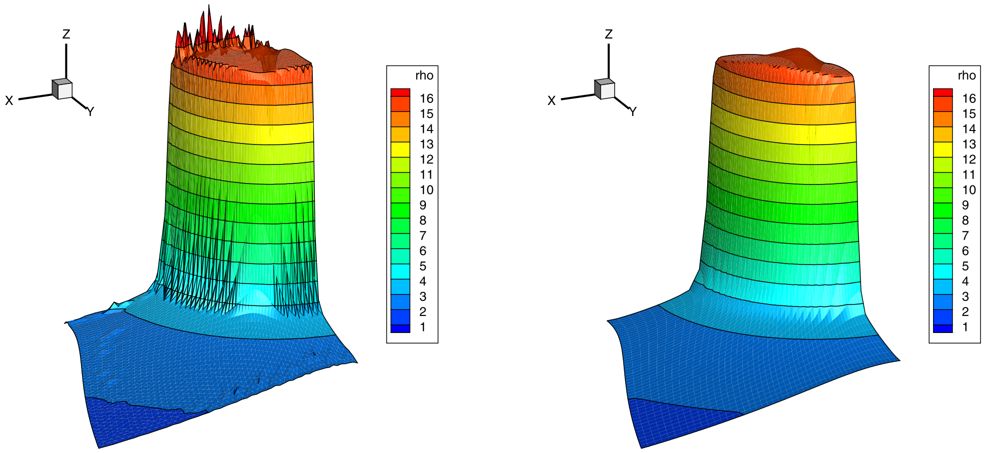


**Fig. 28.** The density contours using the Lagrangian HLLC-2D approach in the Noh problem at time $t = 0.6$. Left: the triangular mesh, right: the quadrilateral mesh.

### 6.8. *Noh test case*

The Noh test case [42] used here is an implosion of a cylinder of unit radius. It has an infinite strong shock and the initial state is

$$(\rho, p, \mathbf{u}, \gamma) = \left(1, 10^{-6}, 0, -\mathbf{e}_R, 5/3\right),$$

where $\mathbf{e}_R$ represents the radial outward unit vector.

In the following, we calculate the test case using the Lagrangian HLLC-2D solver on a square domain $\Omega = [0, 1] \times [0, 1]$ and use both triangular and quadrilateral meshes. The quadrilateral mesh adopts $50 \times 50$ cells, and the Delaunay triangular mesh with similar mesh size has 5856 cells and 3029 nodes.

In Fig. 27 we have represented the grids at $t = 0.6$. The shock is traveling radially outward. Fig. 28 displays the density contours on the two meshes. Because of the triangular mesh includes more cells, the density behind the shock has better accuracy than that on the quadrilateral mesh. However, on the triangular mesh, some oscillations arise near the $x$ axis. These oscillations are caused by irregular triangle distribution of the initial mesh, see Fig. 29. In fact, when we let the initial mesh is generated by another way that the irregular triangles appear near the $y$ axis, the oscillations appear near the $y$ axis. How to overcome such errors on the unstructured mesh is still open. On the other hand, the symmetry of the numerical solution on the quadrilateral mesh is kept rather well.

### 6.9. *Triple point problem*

Finally we consider a two-material triple point problem which corresponds to a three-state two-dimensional Riemann problem. The computational domain $\Omega = [0, 7] \times [0, 3]$ is split into three regions, and the regions and states at the initial

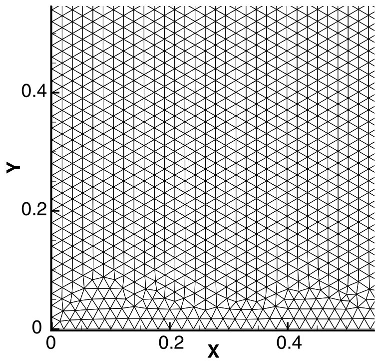


**Fig. 29.** The enlarged initial mesh near the origin in the Noh problem. Some irregular triangles appear near the $x$ axis.

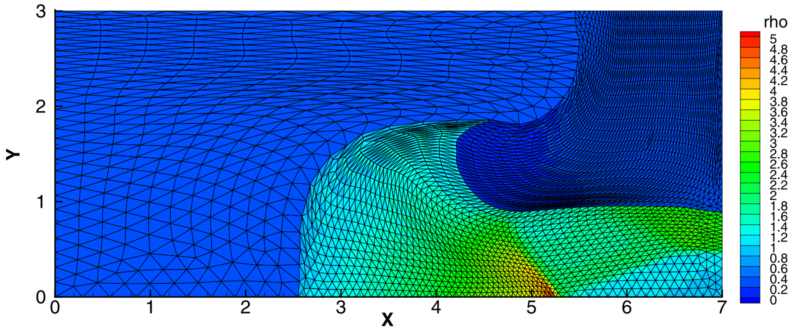


**Fig. 30.** The density field and grid of the second order Lagrangian HLLC-2D approach on a triangular mesh in triple point problem, $t = 5$.

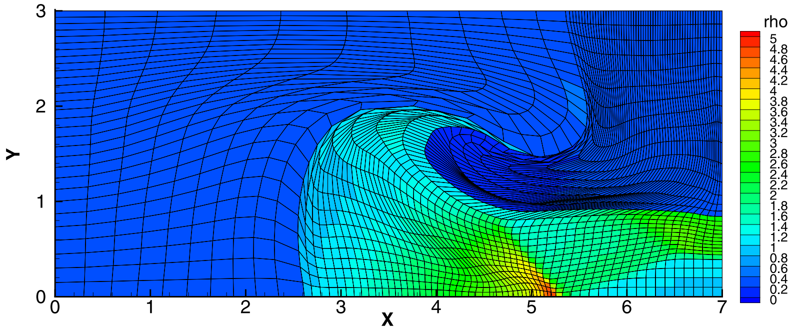


**Fig. 31.** The density field and grid of the second order Lagrangian HLLC-2D approach on a quadrateral mesh in triple point problem, $t = 5$.

time are

$$\Omega_1 = [0,1] \times [0,3]: \quad \rho_1 = 1,\ u_1 = 0,\ v_1 = 0,\ p_1 = 1,\ \gamma_1 = 1.5,$$

$$\Omega_2 = [1,7] \times [0,1.5]: \quad \rho_2 = 1,\ u_2 = 0,\ v_2 = 0,\ p_2 = 0.1,\ \gamma_2 = 1.4,$$

$$\Omega_3 = [1,7] \times [1.5,3]: \quad \rho_3 = 0.125,\ u_3 = 0,\ v_3 = 0,\ p_3 = 0.1,\ \gamma_3 = 1.5.$$

The triple point problem is used to assess the robustness of a Lagrangian method on a problem that has significant vorticity [30]. The left region $\Omega_1$ has a high pressure that drives a shock through two connected regions $\Omega_2 \cup \Omega_3$ and a vortex develops at the triple point where three regions connect. We use quadrilateral and triangular meshes and the Lagrangian HLLC-2D method to simulate this problems. The grid resolution is $70 \times 30$ on a quadrilateral mesh, and a triangular mesh keeps the similar mesh size. The cell number is 5072 and the vertex number is 2637.

Figs. 30 and 31 show the second order Lagrangian solution using triangular and quadrilateral meshes at $t = 5$. Notice that on the Cartesian grid non-convex cells appear, but this does not pose any problem to the HLLC-2D solver. The calculation can run well beyond $t = 6$. However, due to the stiffness caused by the triangles, the vortex at the triple point has not been developed well on the triangular mesh. The calculation on a quadrilateral mesh performs better. The test illustrates our algorithm can be used on problems with significant mesh deformation.

## 7. Summary

This paper presents a numerical investigation for two popular moving mesh schemes when applied to inviscid, compressible gas flows with strong shocks. When the grid motion is generated by a least square procedure, the ALE Godunov and ALE HLLC methods suffer from numerical shock instability in some testing problems.

We have developed a new cell-centered ALE method to improve the stability of the ALE methods associated with one-dimensional Riemann solvers. This new scheme includes a genuinely two-dimensional Riemann solver HLLC-2D based on node states. Different from those multi-dimensional Riemann solvers which need to distinguish multi-dimensional wave configurations, this new algorithm introduces a nodal particle motion velocity and constructs discontinuous fluxes on each side of the edge. The nodal particle velocity is determined via conservation of mass, momentum and total energy.

To guarantee the compatibility between the edge fluxes and the nodal contact velocity, one side Rankine–Hugoniot relations are used to construct the HLLC-2D fluxes and the condition that the contact pressures on an edge must be same is relaxed. The pressure balance is enforced via the nodal solver and the resulting fluxes have consistency and nodal conservation properties.

The new methods have been used for solving a variety of flow problems including strong shock wave, strong rarefaction wave, hourglass and slip line problems. The preliminary numerical results confirm that the ALE HLLC-2D is able to yield oscillation-free and sharp contact solution. Compared with the traditional ALE Godunov and ALE HLLC methods, the ALE HLLC-2D has remarkable advantages in its robustness and stability. It can reduce the numerical shock instability and can restrain nonphysical grid distortion for some 2D test problems.

Moreover the numerical examples show that the ALE HLLC-2D is able to explicitly track material interfaces as a Lagrangian curve in an effort to ensure the unambiguous separation of different material during the computation.

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## Appendix A. Pakmor et al.'s method

Pakmor et al. [43] regard that some unstable behaviors in MHD calculation can be avoided by a kind of moving frame method. However, in the circumstances of gas hydrodynamics, we can show that the flux they proposed is completely equivalent to one-dimensional Riemann solver after some simple algebra manipulation.

Their solution procedure includes two phases. The approximate HLLC Riemann solver is used as an example to describe the procedure.

The first phase is to rewrite the state $\mathbf{U}$ and flux $\mathbf{F}$ in a moving frame as those in a fixed frame. That is

$$\mathbf{U}' = \begin{pmatrix} \rho \\ \rho(u-w) \\ \rho v \\ \rho E' \end{pmatrix}, \qquad \mathbf{F}(\mathbf{U}') = \begin{pmatrix} \rho(u-w) \\ \rho(u-w)^2 + p \\ \rho(u-w)v \\ \rho E'(u-w) + p(u-w) \end{pmatrix},$$

where $E' = e + (u-w)^2/2 = E - uw + w^2/2$.

The left and right states $\mathbf{U}'_L, \mathbf{U}'_R$ can be defined. According to the flux formula (18) of the HLLC solver in a fixed frame, i.e., $w$ in the expression (18) is 0, there holds

$$\mathbf{F}'_{HLLC} = \frac{1}{2}\left[\mathbf{F}(\mathbf{U}'_L) + \mathbf{F}(\mathbf{U}'_R) - \left(\left|S'_L\right|\left(\mathbf{U}'^*_L - \mathbf{U}'_L\right) + \left|S'_*\right|\left(\mathbf{U}'^*_R - \mathbf{U}'^*_L\right) + \left|S'_R\right|\left(\mathbf{U}'_R - \mathbf{U}'^*_R\right)\right)\right]$$

where $\mathbf{U}'^*_K$ for $K = L, R$ satisfies

$$\mathbf{U}'^*_K = \rho_K \frac{S_K - u_K}{S_K - S_*} \begin{bmatrix} 1 \\ S'_* \\ v_K \\ E'_K + (S_* - u_K)(S'_* + \frac{p_K}{\rho_K(S_K - u_K)}) \end{bmatrix}, \tag{A.1}$$

and the wave velocities are $S'_L = S_L - w$, $S'_R = S_R - w$, $S'_* = S_* - w$.

In the second phase, the resulting flux in the moving frame is resumed as

$$\mathbf{F}^c_f(w, \mathbf{U}_L, \mathbf{U}_R) = \mathbf{F}'_{HLLC} + w \begin{pmatrix} 0 \\ \mathbf{F}'_{HLLC}(1) \\ 0 \\ \mathbf{F}'_{HLLC}(2) + w\mathbf{F}'_{HLLC}(1)/2 \end{pmatrix}. \tag{A.2}$$

Substituting $\mathbf{F}'_{HLLC}$ (A.1) into (A.2), the resulting formula (A.2) is the same as the HLLC flux (18).

## Appendix B. Adaptive mesh method on triangular grid

A one-to-one coordinate transformation from the logical or computational domain $\Omega_l$ to the physical domain $\Omega_p$ is denoted by

$$\mathbf{x} = \mathbf{x}(\Pi), \quad \Pi = (\xi, \eta)^T \in \Omega_l, \ \mathbf{x} = (x, y)^T \in \Omega_p.$$

The grid moving strategy is given by solving the following equations

$$\nabla \cdot (W \nabla \mathbf{x}) = 0, \tag{B.1}$$

where $\nabla = (\partial_\xi, \partial_\eta)^T$ is gradient operator in logical space, and the monitor function is

$$W = \sqrt{1 + \alpha_\rho W_c^2(\beta, \rho) + \alpha_p W_c^2(\beta, p)}, \tag{B.2}$$

where $\beta \in (0, 1]$, $\alpha_\rho$ and $\alpha_p$ are some problem-dependent positive parameters, and $W_c$ is the monitor function related to cell $V_c$

$$W_c(\beta, q) = \min\left\{1, \frac{|\nabla q|_c}{\beta \max_c |\nabla q|_c}\right\}, \quad q = \rho, p. \tag{B.3}$$

In practical computation, logical grid is set as initial grid configuration. For more details, refer to [14].

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